Talk transcript: Building quantum codes with transversal CCZ

Virginia Tech Applied Algebra Seminar, October 22, 2026. Joint work with Ken Brown. About 38 minutes, then discussion.

How to use this transcript

Each section is one frame of the deck, in order. The heading gives the displayed slide number. The five unnumbered outline frames between parts have their own sections.

- **Clock** is the planned start and end time. **Budget** is the time for the frame. **Clicks** is the number of overlay steps after the first view.
- **[click]** marks where to advance. The text after it describes what has just appeared.
- *(pause)* means stop for about two seconds and let the slide be read.
- *(ask, then wait 7 seconds)* marks a question to the audience. A fallback answer follows in case nobody answers.
- Spoken text is written at about 135 words per minute. Read it aloud at a steady pace with the clock running, and mark where you run over.
- **Background (not spoken)** boxes on slides 2 to 8 and on a few later slides collect the facts behind each sentence, with short answers to likely questions. Read them before rehearsing, not during.

Terms and notation follow the standard literature and our paper: Pauli strings $X(a)$ and $Z(b)$, stabilizers and stabilizer generators, X-type and Z-type stabilizers, logical operators, the CSS code of $\mathcal{C}_2 \subseteq \mathcal{C}_1$, quasi-transversal, triorthogonal, column indicator. Two terms come from our paper and are defined on the slides: the signed weight and the logical label.

Winston's "How to Speak" applied to this talk

These heuristics are from Patrick Winston's MIT lecture "How to Speak" (MIT OpenCourseWare RES.TLL-005, IAP 2018). Each one is built into the script.

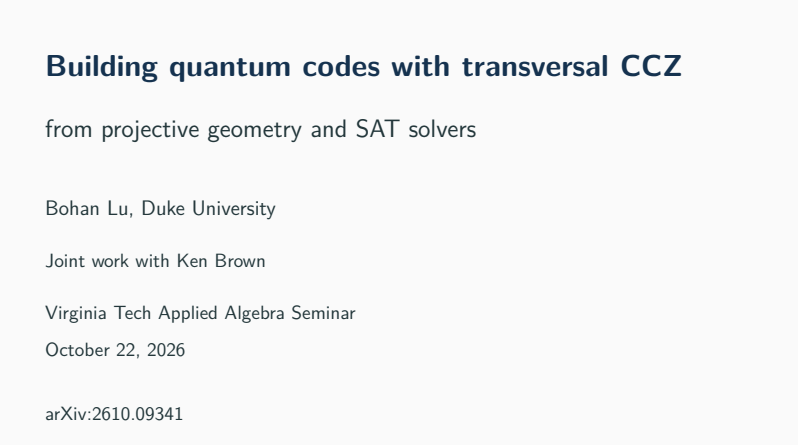
- **Empowerment promise, not a joke.** Slide 1 tells the audience what they will be able to do at the end, before 1:00. The result itself waits for slide 15, after the background that lets the audience follow it. Winston says a speaker should show both a vision and that something has been done within five minutes. The promise is the vision, and the Steane code on slides 5 to 8 and the $[\![8, 3, 2]\!]$ code on slides 9 to 12 show that something has been done.
- **Cycling.** The three steps of the construction are said at slides 15, 16 and 29: "Geometry fixes the stabilizers S, a search finds the logical rows K, and linear algebra over $\mathbb{Z}_4$ finds the signs."
- **Fences.** Each of these is said once, at the slide where it applies:
  - quasi-transversal CCZ versus a $T/T^\dagger$ pattern with no correction (slides 14 and 28)
  - the Nezami–Haah classification range $n + k \leq 38$ versus our bound $n \leq 38$ (slide 13)
  - 8-divisibility is a design choice, not part of the conditions (slide 16)
  - the 47-qubit code of Jacinto et al. is not $\mathcal{Q}_{48}$ with one point removed (slide 28).
- **Verbal punctuation.** Each outline frame says which part of six begins and what it does.
- **Ask a question.** Two planned questions, at slides 11 and 19, each with a 7-second wait.
- **Collaborators first.** Ken Brown and Robert Calderbank are credited on the title slide.
- **Ending.** Slide 29 is read as contributions, and the talk ends with a salute to the audience instead of "thank you".

Pacing checkpoints

Reach this point	Clock	If about one minute behind
Outline before slide 9	7:50	On slide 9, show the $x = 100$ pair and name the $x = 111$ pair without a pause.
Outline before slide 13	13:00	Give the timeline in two sentences. Keep the odd-triple point and the parity/sign distinction.
Outline before slide 21	22:40	On slide 20, give the formula, the Q48 entry and the census qualification only.
Outline before slide 26	30:40	Give the Reed–Muller argument in two sentences. Name the four stages of the bound without detail.
Slide 29	36:10	Read the contributions and two open questions. Stop at 37:40.

Do not save time by dropping the sign-feasibility distinction or the hypotheses of the lower bound. If a question interrupts a derivation, give a short answer and offer to return to it in the discussion.

Title slide

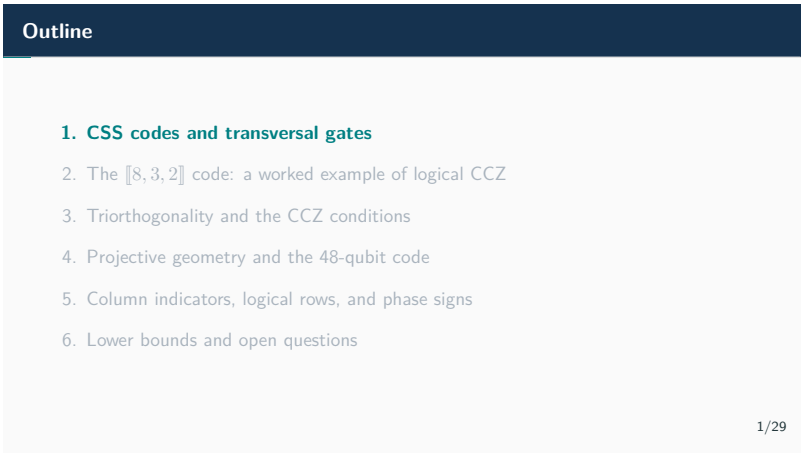


Clock 0:00–0:15 · **Budget** 15 s · **Clicks** 0

On screen: title, Bohan Lu and Ken Brown, arXiv number.

Thank you for the invitation. This is joint work with Ken Brown at Duke, and I also thank Robert Calderbank for pointers on the classical coding theory.

Slide 1. Outline

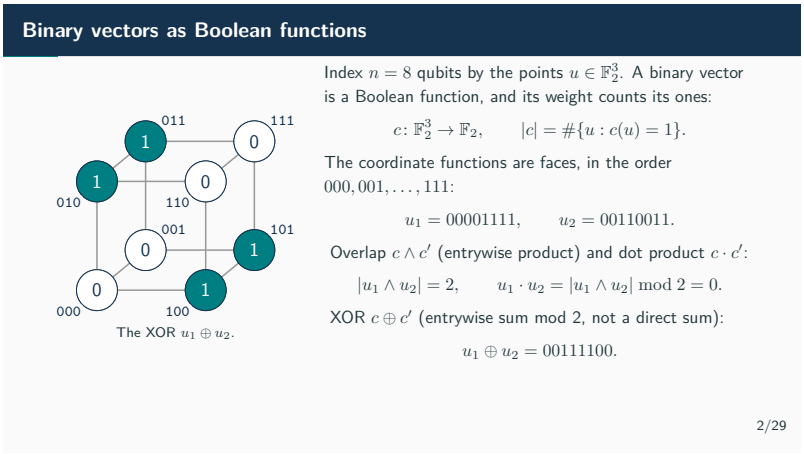


Clock 0:15–1:00 · **Budget** 45 s · **Clicks** 0

On screen: the six parts of the talk, with part 1 highlighted.

By the end of this talk you will be able to take a binary generator matrix and a choice of T or $T^\dagger$ on each qubit, and decide whether this pattern of gates acts as a logical CCZ. *(pause)* We start with the quantum background we need, then work through an eight-qubit example in full. After that comes the construction of a larger code, and finally the lower bounds.

Slide 2. Binary vectors as Boolean functions



Clock 1:00–1:55 · **Budget** 55 s · **Clicks** 4

On screen: the cube $\mathbb{F}_2^3$ with one qubit at each point, and the weight $|c|$ of a Boolean function. The clicks show u_1, u_2, the overlap $u_1 \wedge u_2$ and the XOR $u_1 \oplus u_2$ on the cube.

The cube $\mathbb{F}_2^3$ is our drawing board. Put one qubit at each of its eight points u. A binary vector c of length eight is then a Boolean function on the points, and its weight $|c|$ counts the points where it is one.

[click] The coordinate function u_1 is one on the face where the first coordinate is one. In the order 000 to 111 it is 00001111, of weight four.

[click] u_2 is the face in the second direction.

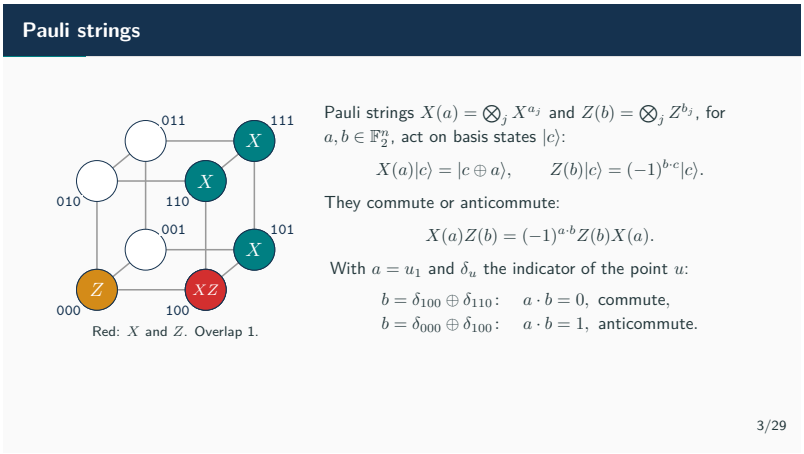
[click] The overlap $u_1 \wedge u_2$ is the entrywise product, one where both are one. Here it is an edge, of weight two. The dot product $u_1 \cdot u_2$ is the parity of the overlap, so it is zero.

[click] The XOR $u_1 \oplus u_2$ is entrywise addition mod 2. It is XOR, not a direct sum. These definitions hold for any length n.

Background (not spoken)

- **Dot product versus overlap.** $c \cdot c'$ is a single bit, $\sum_i c_i c'_i \bmod 2$. The overlap $c \wedge c'$ is a vector. They are related by $c \cdot c' = |c \wedge c'| \bmod 2$. Later slides use the overlap and its integer weight for signed weights, and the dot product only as a sign exponent or a parity.
- **Why the cube.** The cube is only a picture of $\mathbb{F}_2^3$. Faces are the sets $u_i = 1$, two faces meet in an edge, and three faces meet in a vertex. The 48-qubit code later uses the same idea on 48 points of $\mathbb{F}_2^6$.

Slide 3. Pauli strings



Clock 1:55–2:55 · **Budget** 60 s · **Clicks** 3

On screen: the Pauli strings $X(a)$ and $Z(b)$ and their action on $|c\rangle$, with $X(u_1)$ on the cube. The clicks add the commutation rule and two examples.

For binary vectors a and b, the Pauli string $X(a)$ puts an X on each qubit where a is one, and $Z(b)$ puts a Z where b is one. On the left is $X(u_1)$, an X on each point of the face. On a basis state, $X(a)$ sends $|c\rangle$ to $|c \oplus a\rangle$, and $Z(b)$ multiplies $|c\rangle$ by $(-1)^{b\cdot c}$.

[click] Pauli strings commute or anticommute: $X(a)Z(b) = (-1)^{a\cdot b}Z(b)X(a)$. Each qubit in the overlap contributes one minus sign.

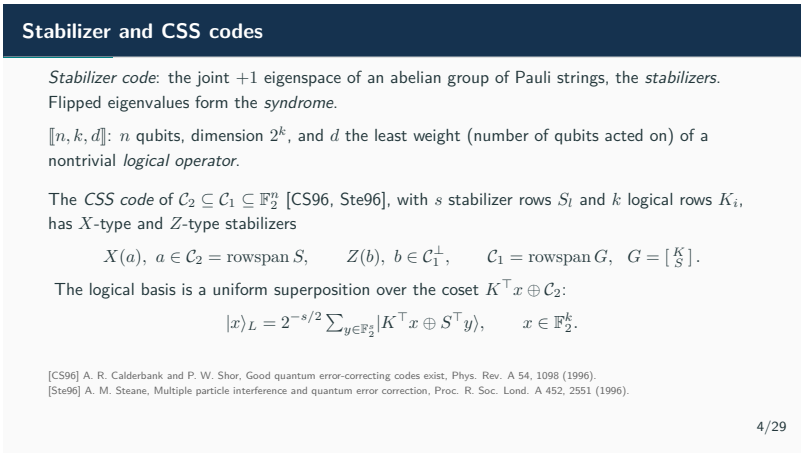
[click] Take $a = u_1$, and write δ_u for the indicator of the point u. Put Z on the edge from 100 to 110. Both points are in the face, shown in red, so $a \cdot b = 0$ and the strings commute.

[click] Now put Z on the edge from 000 to 100. Only 100 is in the face, so $a \cdot b = 1$ and they anticommute.

Background (not spoken)

- **Notation.** $X(a)$ and $Z(b)$ with $a, b \in \mathbb{F}_2^n$ is the notation of Camps-Moreno et al. and of Rengaswamy, Calderbank and coauthors. The Pauli group is generated by $\pm X(a)Z(b)$, with $Y = iXZ$ on a single qubit. For general strings, $X(a)Z(b)$ and $X(a')Z(b')$ commute up to the sign $(-1)^{a\cdot b' + b\cdot a'}$, the symplectic inner product. The talk only needs the case $X(a)$ against $Z(b)$.
- **Why Pauli strings suffice.** I, X, Z and XZ span the 2×2 complex matrices. The Knill–Laflamme conditions are linear in the error, so a code that corrects a set of Pauli errors also corrects every linear combination of them.

Slide 4. Stabilizer and CSS codes



Stabilizer and CSS codes

Stabilizer code: the joint $+1$ eigenspace of an abelian group of Pauli strings, the *stabilizers*. Flipped eigenvalues form the *syndrome*.

$[\![n, k, d]\!]$: n qubits, dimension 2^k, and d the least weight (number of qubits acted on) of a nontrivial *logical operator*.

The *CSS code* of $\mathcal{C}_2 \subseteq \mathcal{C}_1 \subseteq \mathbb{F}_2^n$ [CS96, Ste96], with s stabilizer rows S_l and k logical rows K_i, has X-type and Z-type stabilizers

$$X(a),\ a \in \mathcal{C}_2 = \operatorname{rowspan} S, \qquad Z(b),\ b \in \mathcal{C}_1^{\perp}, \qquad \mathcal{C}_1 = \operatorname{rowspan} G,\quad G = \left[\begin{smallmatrix} K \\ S \end{smallmatrix}\right].$$

The logical basis is a uniform superposition over the coset $K^{\top}x \oplus \mathcal{C}_2$:

$$|x\rangle_L = 2^{-s/2} \textstyle\sum_{y \in \mathbb{F}_2^s} |K^{\top}x \oplus S^{\top}y\rangle, \qquad x \in \mathbb{F}_2^k.$$

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
[Ste96] A. M. Steane, Multiple particle interference and quantum error correction, Proc. R. Soc. Lond. A 452, 2551 (1996).

4/29

Clock 2:55–3:55 · **Budget** 60 s · **Clicks** 2

On screen: stabilizer code, syndrome and $[\![n, k, d]\!]$. The clicks add the CSS code of $\mathcal{C}_2 \subseteq \mathcal{C}_1$ with S, K and G, and the logical basis $|x\rangle_L$.

A stabilizer code is the joint $+1$ eigenspace of an abelian group of Pauli strings, the stabilizers. An error that anticommutes with a stabilizer flips its eigenvalue, and the flipped eigenvalues form the syndrome. An $[\![n, k, d]\!]$ code has n qubits and dimension 2^k. d is the least weight of a nontrivial logical operator, a Pauli string that commutes with every stabilizer but is not one.

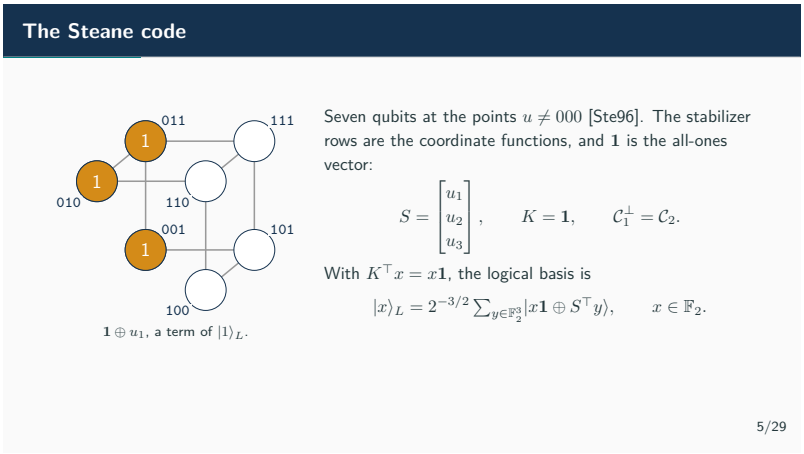
[click] A CSS code comes from two nested binary codes, $\mathcal{C}_2 \subseteq \mathcal{C}_1$. The X-type stabilizers are $X(a)$ for $a \in \mathcal{C}_2$, and the Z-type stabilizers are $Z(b)$ for $b \in \mathcal{C}_1^{\perp}$. $\mathcal{C}_2$ is spanned by the s stabilizer rows of S, and $\mathcal{C}_1$ by G, the k logical rows of K stacked on S.

[click] The logical basis state $|x\rangle_L$ is the uniform superposition over the coset $K^{\top}x \oplus \mathcal{C}_2$. The next slide fills in each symbol for the Steane code.

Background (not spoken)

- **Why an anticommuting error flips the eigenvalue.** If $P|\psi\rangle = |\psi\rangle$ and $PE = -EP$, then $P(E|\psi\rangle) = -E|\psi\rangle$. So the corrupted state is in the -1 eigenspace of P.
- **Why eigenvalues are ± 1.** Each stabilizer is a Hermitian Pauli string with $P^2 = I$. The group must not contain $-I$.
- **Weight.** The weight of $X(a)Z(b)$ is the number of qubits where a or b is one.
- **Why $X(a)$ with $a \in \mathcal{C}_2$ fixes $|x\rangle_L$.** $X(a)$ sends $|K^{\top}x \oplus S^{\top}y\rangle$ to another term of the same sum, so it only permutes the terms.
- **Why $Z(b)$ with $b \in \mathcal{C}_1^{\perp}$ fixes $|x\rangle_L$.** $Z(b)|c\rangle = (-1)^{b \cdot c}|c\rangle$, and $b \cdot c = 0$ for every $c \in \mathcal{C}_1$.
- **Why they commute.** $X(a)Z(b) = (-1)^{a \cdot b}Z(b)X(a)$, and $a \cdot b = 0$ because $a \in \mathcal{C}_2 \subseteq \mathcal{C}_1$ and $b \in \mathcal{C}_1^{\perp}$.
- **Counting.** $k = \dim \mathcal{C}_1 - \dim \mathcal{C}_2$. There are s X-type generators and $n - k - s$ Z-type generators, $n - k$ in all.
- **Notation in the literature.** Nielsen and Chuang, Camps-Moreno et al. and our paper write the CSS code of $\mathcal{C}_2 \subseteq \mathcal{C}_1$ as here. Our paper writes the logical state as $K^{\top}x + S^{\top}y$. The slides write the same vector as $K^{\top}x \oplus S^{\top}y$, with $\oplus$ for addition mod 2, to keep it apart from the integer sums in signed weights.
- **If asked "is every stabilizer code CSS?"** No. In a CSS code every stabilizer generator is all X or all Z. All codes in this talk are CSS.

Slide 5. The Steane code



Clock 3:55–4:35 · **Budget** 40 s · **Clicks** 2

On screen: the seven nonzero points of the cube, $S = [u_1; u_2; u_3]$, $K = \mathbf{1}$, and $X(u_1)$ on the cube. The clicks add the logical basis $|x\rangle_L$, with one term of $|0\rangle_L$ and one term of $|1\rangle_L$ on the cube.

Here is the first code. Remove the origin and keep the seven qubits at the nonzero points. Take S with rows u_1, u_2, u_3, so the X-type stabilizers sit on faces. On the left is $X(u_1)$. Take one logical row, $K = \mathbf{1}$, the all-ones vector. Then $\mathcal{C}_1^\perp = \mathcal{C}_2$, so the Z-type stabilizers sit on the same faces.

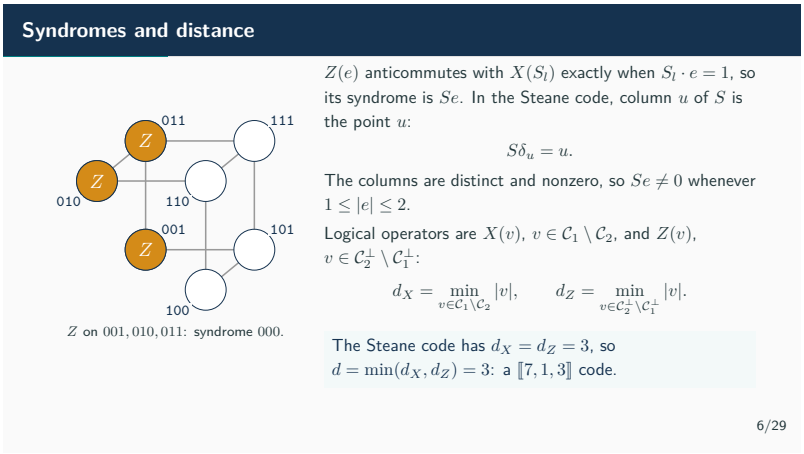
[click] Here $K^\top x = x\mathbf{1}$. So the terms of $|0\rangle_L$ are the codewords of $\mathcal{C}_2$, for example the face u_1.

[click] The terms of $|1\rangle_L$ are their complements, for example $\mathbf{1} \oplus u_1$.

Background (not spoken)

- **Classical codes.** $\mathcal{C}_2$ is the $[7, 3, 4]$ simplex code and $\mathcal{C}_1 = \mathcal{C}_2 \oplus \langle \mathbf{1} \rangle$ is the $[7, 4, 3]$ Hamming code. $\mathcal{C}_1$ is RM$(1, 3)$ with the origin deleted (punctured).
- **Weights.** The terms of $|0\rangle_L$ have weights 0 and 4, and the terms of $|1\rangle_L$ have weights 7 and 3.
- **Fano plane.** The seven nonzero points of $\mathbb{F}_2^3$ are the points of PG$(2, 2)$. Its seven lines $\{u, v, u \oplus v\}$ are the weight-3 codewords of $\mathcal{C}_1$.

Slide 6. Syndromes and distance



Clock 4:35–5:40 · **Budget** 65 s · **Clicks** 2

On screen: the syndrome Se of $Z(e)$ and $S\delta_u = u$, with Z on 101. The clicks add two errors, then the logical operators, d_X and d_Z, and Z on a weight-3 line.

$Z(e)$ anticommutes with $X(S_l)$ exactly when $S_l \cdot e = 1$, so the syndrome of $Z(e)$ is the vector Se. For the Steane code, column u of S is the point u itself. So Z on qubit 101 has syndrome 101: the syndrome names the qubit.

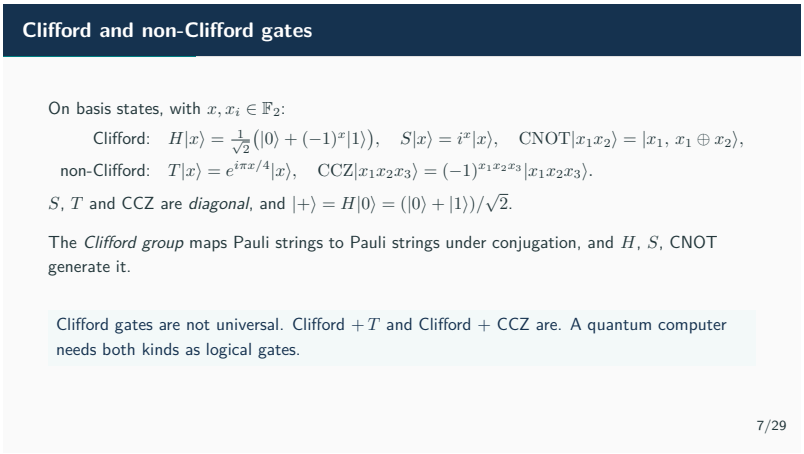
[click] Z on 101 and 011 has syndrome $101 \oplus 011 = 110$. Because the columns are distinct and nonzero, every Z error of weight one or two is detected.

[click] The logical operators are $X(v)$ for v in $\mathcal{C}_1$ but not in $\mathcal{C}_2$, and $Z(v)$ for v in $\mathcal{C}_2^\perp$ but not in $\mathcal{C}_1^\perp$. Their minimum weights are d_X and d_Z. Here Z on 001, 010 and 011 has syndrome 000. It has odd weight, so it is not a stabilizer. It is a logical operator of weight three, and the Steane code is a $[\![7, 1, 3]\!]$ code.

Background (not spoken)

- **Decoding.** $Z(e)$ is undetected exactly when $e \in \mathcal{C}_2^\perp$, and $X(e)$ exactly when $e \in \mathcal{C}_1$. So Z errors are corrected by decoding $\mathcal{C}_2^\perp$, and X errors by decoding $\mathcal{C}_1$.
- **Logical Z.** For $b \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp$, $Z(b)$ acts on $|x\rangle_L$ as the sign $(-1)^{b\cdot(K^\top x)}$, which depends on x. That is why it is a nontrivial logical operator.
- **d_X.** By symmetry ($\mathcal{C}_1^\perp = \mathcal{C}_2$) the same line gives an X logical of weight three.
- **Later use.** The 48-qubit code uses the same argument: distinct nonzero columns of S give $d_Z \geq 3$.

Slide 7. Clifford and non-Clifford gates



Clock 5:40–6:40 · **Budget** 60 s · **Clicks** 2

On screen: H, S, CNOT, T and CCZ, each written by its action on basis states, and the definition of a diagonal gate.

Now gates. Each gate is written by what it does to a basis state $|x\rangle$, with x a bit. H sends $|x\rangle$ to $|0\rangle+(-1)^x|1\rangle$, divided by $\sqrt{2}$. The phase gate S, not the matrix S, multiplies $|x\rangle$ by i^x. CNOT adds the first bit to the second. T multiplies $|x\rangle$ by $e^{i\pi x/4}$. CCZ multiplies $|x_1x_2x_3\rangle$ by $(-1)^{x_1x_2x_3}$, so it puts a minus sign on $|111\rangle$ and nothing else. S, T and CCZ are diagonal. We write $|+\rangle$ for $H|0\rangle$.

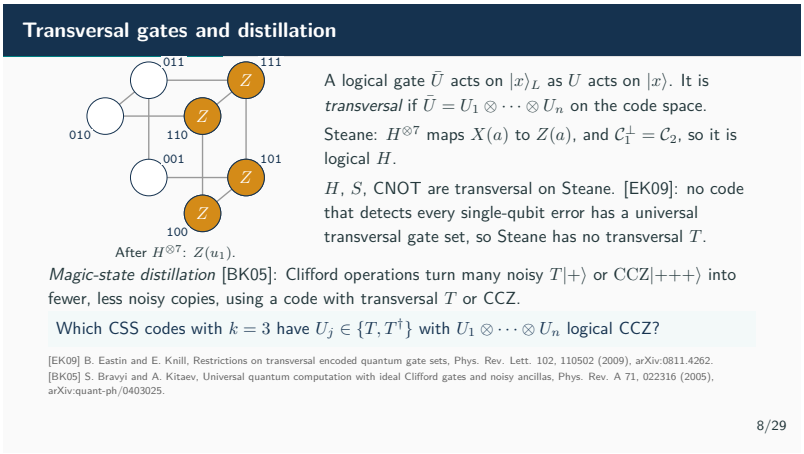
[click] The Clifford group maps Pauli strings to Pauli strings under conjugation. H, S and CNOT generate it. That is the first row. T and CCZ, in the second row, are not Clifford.

[click] Clifford gates alone are not universal. Adding T, or adding CCZ, makes the gate set universal. So a quantum computer needs both kinds as logical gates.

Background (not spoken)

- **Clifford group.** The unitaries U with $UPU^\dagger$ a Pauli string, up to phase, for every Pauli string P. For example, $HXH = Z$, $SXS^\dagger = Y$ with $Y = iXZ$, and CNOT maps $X \otimes I$ to $X \otimes X$.
- **Why T is not Clifford.** $TXT^\dagger = (X + Y)/\sqrt{2}$, which is not a Pauli string up to phase. For CCZ, conjugating X on the first qubit gives X times a controlled-Z on the other two, which is not a Pauli string.
- **Why Clifford circuits are not universal.** By the Gottesman–Knill theorem, a Clifford circuit on stabilizer states with Pauli measurements can be simulated efficiently on a classical computer.
- **Universality.** Clifford + T generates a dense subgroup of the unitary group up to phase. CCZ with H gives Toffoli, and Clifford + CCZ is also universal.
- **CNOT convention.** $|x_1, x_1 \oplus x_2\rangle$: the first qubit is the control.
- **If asked "why CCZ rather than T?"** The standard Clifford + T circuit for CCZ uses seven T gates. Campbell and Howard distill CCZ states directly, which can lower the cost.

Slide 8. Transversal gates and distillation



Clock 6:40–7:50 · **Budget** 70 s · **Clicks** 4

On screen: the definition of a logical gate and of a transversal gate, with $X(u_1)$ on the Steane cube. The clicks show $H^{\otimes 7}$ turning it into $Z(u_1)$, then Eastin–Knill, then distillation, then the question of the talk.

A logical gate $\bar{U}$ acts on $|x\rangle_L$ as U acts on $|x\rangle$. It is transversal if it is a product of one gate on each physical qubit, $U_1 \otimes \cdots \otimes U_n$.

[click] Put H on every qubit of the Steane code. $H^{\otimes 7}$ maps $X(a)$ to $Z(a)$: the X on the face becomes a Z on the same face. Because $\mathcal{C}_1^\perp = \mathcal{C}_2$, that is again a stabilizer, and $H^{\otimes 7}$ acts as logical H.

[click] H, S and CNOT are all transversal on the Steane code. Eastin and Knill proved that no code that detects every single-qubit error has a universal transversal gate set. So the Steane code has no transversal T.

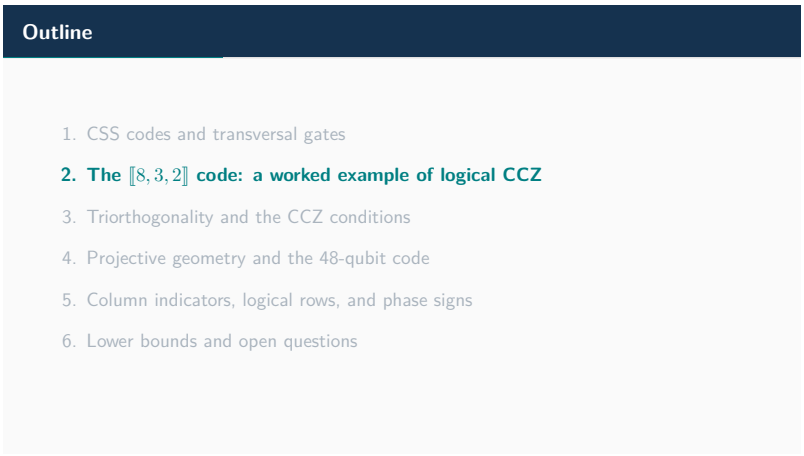
[click] So at least one gate needs another method. Magic-state distillation, from Bravyi and Kitaev, turns many noisy copies of $T|+\rangle$ or $\mathrm{CCZ}|{+}{+}{+}\rangle$ into fewer, less noisy copies, using a code with transversal T or CCZ.

[click] So here is the question of this talk. *(pause)* For which CSS codes with three logical qubits can we put T or $T^\dagger$ on each qubit so that the product is logical CCZ?

Background (not spoken)

- **Logical H in detail.** $H^{\otimes 7}$ also swaps the logical operators $X(\mathbf{1})$ and $Z(\mathbf{1})$, which is what logical H does.
- **Transversal S.** $S^{\otimes 7}$ multiplies $|c\rangle$ by $i^{|c|}$. With weights $0, 4$ in $|0\rangle_L$ and $7, 3$ in $|1\rangle_L$, it is logical $S^\dagger$, and $(S^\dagger)^{\otimes 7}$ is logical S.
- **CNOT.** CNOT between corresponding qubits of two blocks is logical CNOT for every CSS code.
- **The idea behind Eastin–Knill.** The transversal logical gates form a group. If the code detects every single-qubit error, a single-qubit unitary close to the identity acts trivially on the code space. So the group of transversal logical gates is finite, and a finite group cannot be dense in the unitary group.
- **Magic states.** $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $|T\rangle = T|+\rangle$ and $|\mathrm{CCZ}\rangle = \mathrm{CCZ}|{+}{+}{+}\rangle$. Gate teleportation consumes one such state, together with Clifford gates and measurement, to apply T or CCZ.
- **If asked for an example protocol.** The 15-to-1 protocol of Bravyi and Kitaev applies T on every qubit of a 15-qubit Reed–Muller code, on which it acts as logical $T^\dagger$.
- **Why transversal.** A single faulty gate damages only one qubit of the block.
- **If asked about "single-qubit".** The gates U_j may differ from qubit to qubit. The $[\![8, 3, 2]\!]$ code uses T on some qubits and $T^\dagger$ on others.

Outline before slide 9 (part 2 of 6)

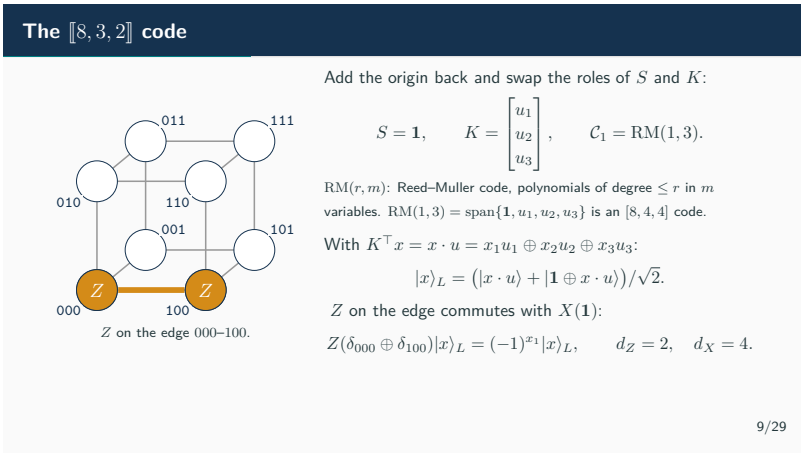


Clock 7:50–8:00 · **Budget** 10 s · **Clicks** 0

On screen: the outline, with part 2 highlighted.

Part two answers this question for the $[\![8,3,2]\!]$ code on the same cube: its encoding, its phase, and the overlap conditions for CCZ.

Slide 9. The $[\![8, 3, 2]\!]$ code



Clock 8:00–9:15 · **Budget** 75 s · **Clicks** 3

On screen: $S = \mathbf{1}$, $K = [u_1; u_2; u_3]$, $\mathcal{C}_1 = \mathrm{RM}(1,3)$, and $X(\mathbf{1})$ on the cube. The clicks add the logical basis state, show its two terms for $x = 100$ and for $x = 111$, then a weight-two Z logical on an edge.

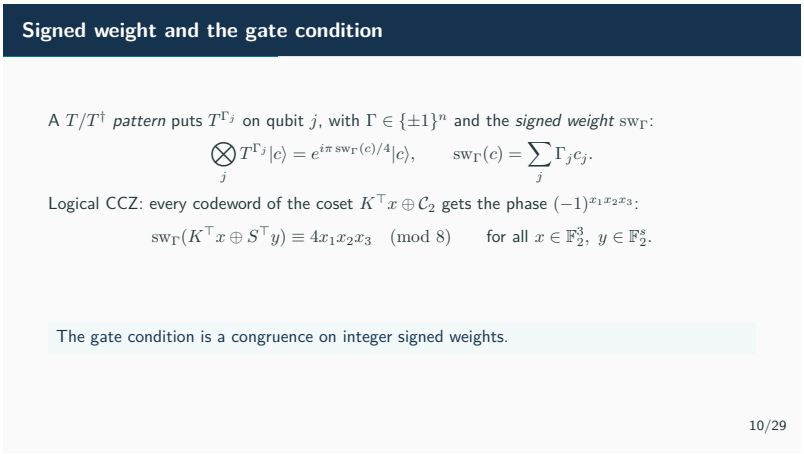
Put the origin back, so all eight points are qubits, and swap the roles of S and K. Now $S = \mathbf{1}$, so the only X-type stabilizer generator is $X(\mathbf{1})$, on the left. K has the three faces u_1, u_2, u_3 as rows. Then $\mathcal{C}_1$ is the Reed–Muller code RM(1, 3) of polynomials of degree at most one, an $[8, 4, 4]$ code, and $k = 3$.

[click] With $K^\top x = x \cdot u$, each logical state is a pair: $x \cdot u$ and its complement. For $x = 100$ the pair is the face u_1, in teal, and the opposite face, in gold.

[click] For $x = 111$ the pair is $u_1 \oplus u_2 \oplus u_3$ and its complement. This is the label that CCZ multiplies by minus one.

[click] How many Z errors does it take to change a logical state without being detected? Two: put Z on the edge from 000 to 100. The edge has even overlap with $\mathbf{1}$, so it commutes with $X(\mathbf{1})$, and it multiplies $|x\rangle_L$ by $(-1)^{x_1}$. So $d_Z = 2$. On the X side, RM(1, 3) has minimum weight four, so $d_X = 4$. *(pause)* The code has distance two. We want distance three.

Slide 10. Signed weight and the gate condition



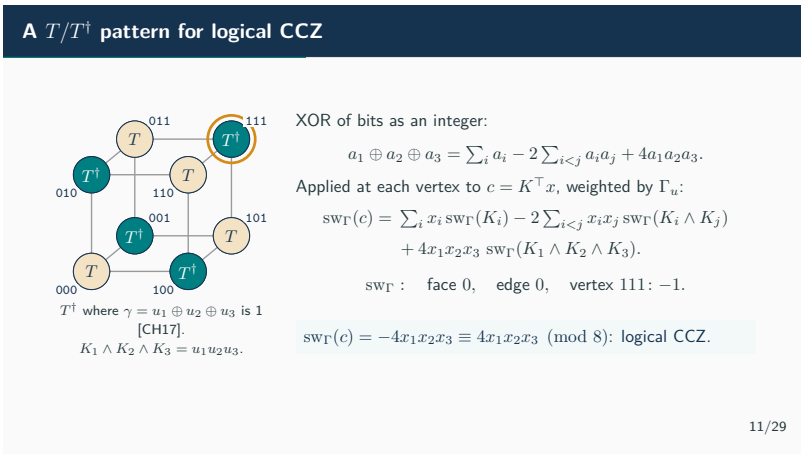
Clock 9:15–10:15 · **Budget** 60 s · **Clicks** 1

On screen: the phase of a $T/T^\dagger$ pattern on a basis state, and the signed weight sw_Γ.

Why does a pattern of T and $T^\dagger$ gates act by a phase? Write the gate on qubit j as T^{Γ_j}, with $\Gamma_j = \pm 1$, so T^{-1} is $T^\dagger$. Each T^{Γ_j} multiplies $|c_j\rangle$ by $e^{i\pi\Gamma_j c_j/4}$. We call the product a $T/T^\dagger$ pattern. It multiplies the basis state $|c\rangle$ by $e^{i\pi \mathrm{sw}_\Gamma(c)/4}$. Here $\mathrm{sw}_\Gamma(c) = \sum_j \Gamma_j c_j$ is the signed weight: the number of ones of c on T qubits minus the number on $T^\dagger$ qubits. It is an integer, not a parity.

[click] The pattern is diagonal, so it multiplies each term of a logical state by that term's own phase. It acts as a logical gate only if all terms of a logical state receive the same phase. For CCZ that phase is $(-1)^{x_1x_2x_3}$. In terms of signed weights, $\mathrm{sw}_\Gamma(K^\top x \oplus S^\top y) \equiv 4x_1x_2x_3 \pmod 8$ for all x and y. This congruence is the gate condition, and the rest of the talk rewrites it.

Slide 11. A $T/T^\dagger$ pattern for logical CCZ



Clock 10:15–12:15 · **Budget** 120 s · **Clicks** 4

On screen: the blank cube, the integer expansion of $a_1 \oplus a_2 \oplus a_3$, and $\mathrm{sw}_\Gamma(c)$ in terms of overlaps. The clicks show the pattern with $T^\dagger$ where $\gamma = u_1 \oplus u_2 \oplus u_3$ is one, then a face, an edge and a vertex.

Now we look for the signs. For three bits, XOR has an integer expansion: $a_1 \oplus a_2 \oplus a_3 = \sum_i a_i - 2\sum_{i<j} a_i a_j + 4a_1a_2a_3$. Apply it at each vertex to the bits of $c = K^\top x$, weight by Γ_u, and sum. That gives $\mathrm{sw}_\Gamma(c)$ in terms of the signed weights of the rows K_i, of their pairwise overlaps, and of the triple overlap. For CCZ, only the last term should survive. *(ask, then wait 7 seconds)* Where would you put the $T^\dagger$ gates? *(Fallback: where $u_1 \oplus u_2 \oplus u_3$ is one.)*

[click] Campbell and Howard put $T^\dagger$ where $\gamma = u_1 \oplus u_2 \oplus u_3$ is one, and T elsewhere.

[click] $K_1 = u_1$ is the face $u_1 = 1$. It has T at 101 and 110, and $T^\dagger$ at 100 and 111. So $\mathrm{sw}_\Gamma(K_1) = 0$, and the same holds for K_2 and K_3.

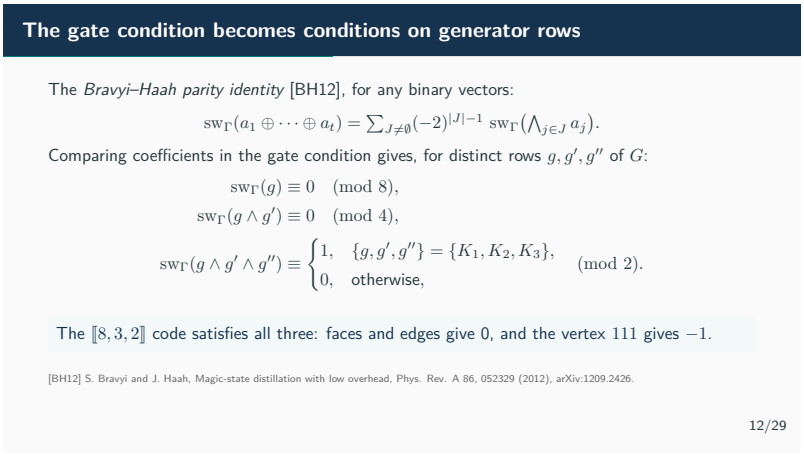
[click] $K_1 \wedge K_2$ is the edge $\{110, 111\}$. That is one T and one $T^\dagger$, so its signed weight is 0, and likewise for the other pairs.

[click] $K_1 \wedge K_2 \wedge K_3$ is the vertex 111, which carries $T^\dagger$. So its signed weight is -1, and $\mathrm{sw}_\Gamma(c) = -4x_1x_2x_3 \equiv 4x_1x_2x_3 \pmod 8$. That is CCZ, read off a face, an edge and a vertex. The stabilizer row **1** adds terms of the same kind, a face and an edge, and they vanish too.

Background (not spoken)

- **A second check.** With $\Gamma_u = (-1)^{\gamma(u)}$, $\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c| = |\gamma \oplus c| - 4$. Every codeword other than γ and its complement is at distance 4 from γ, so it gets phase 1. The two codewords of the label 111 are at distance 0 and 8, so both get phase -1.
- **Source.** This is the eight-qubit example of Campbell and Howard.

Slide 12. The gate condition becomes conditions on generator rows



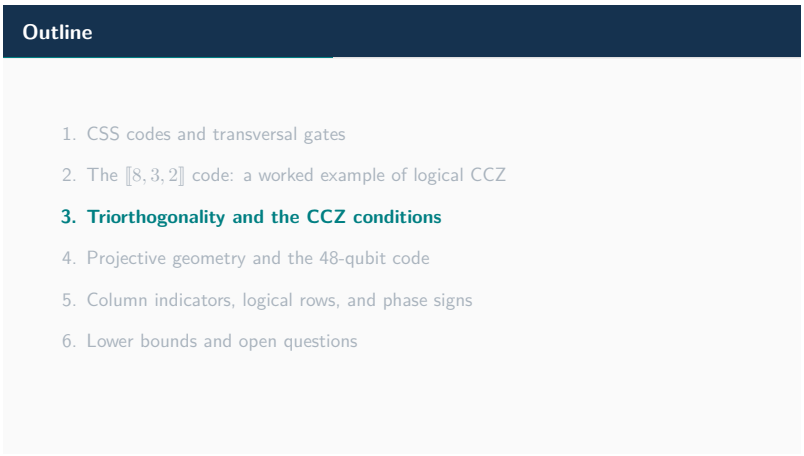
Clock 12:15–13:00 · **Budget** 45 s · **Clicks** 1

On screen: the Bravyi–Haah parity identity, and the three congruences for single rows, pairs and triples.

The same expansion holds for any number of vectors. This is the Bravyi–Haah parity identity: an overlap of t vectors has coefficient $(-2)^{t-1}$. Expand the gate condition this way and compare coefficients. For distinct rows g, g', g'' of G, single rows need $\mathrm{sw}_\Gamma(g) \equiv 0 \pmod 8$. Pair overlaps carry -2, so they need $\mathrm{sw}_\Gamma(g \wedge g') \equiv 0 \pmod 4$. Triple overlaps carry 4, so only their parity matters, and it must be odd exactly for the triple K_1, K_2, K_3. Overlaps of four or more rows carry a multiple of eight and drop out.

[click] The $[\![8, 3, 2]\!]$ code satisfies all three. These are the conditions we impose on larger codes.

Outline before slide 13 (part 3 of 6)

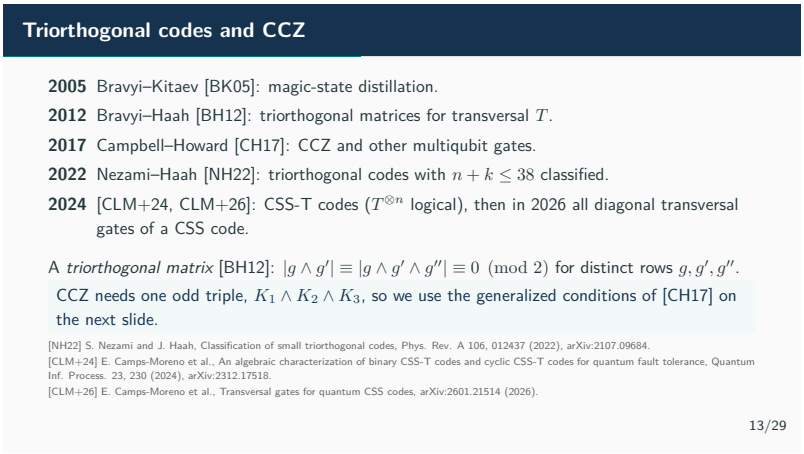


Clock 13:00–13:10 · **Budget** 10 s · **Clicks** 0

On screen: the outline, with part 3 highlighted.

Part three. Now I place these conditions in the literature, and separate parities from signs.

Slide 13. Triorthogonal codes and CCZ



Clock 13:10–14:10 · **Budget** 60 s · **Clicks** 1

On screen: a timeline from 2005 to 2026. The click adds the triorthogonal matrix condition and the need for one odd triple.

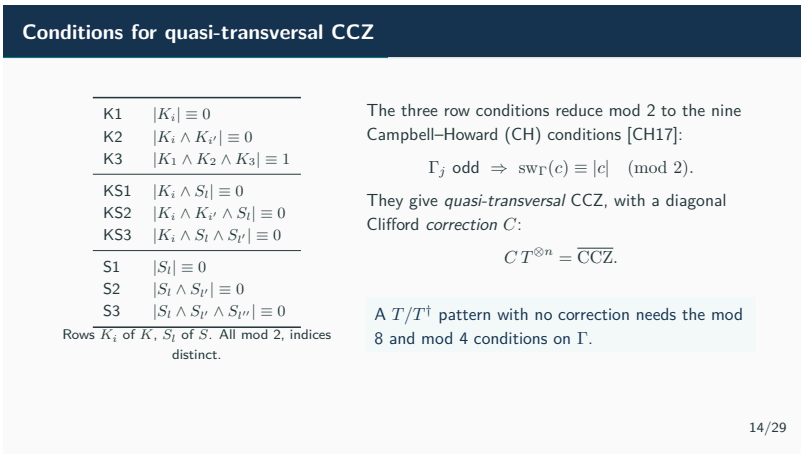
Briefly, the history. Bravyi and Kitaev introduced magic-state distillation. Bravyi and Haah wrote transversal T as parity conditions on binary rows, triorthogonal matrices. Campbell and Howard gave the CCZ version, which we use. Nezami and Haah classified triorthogonal codes with $n + k \leq 38$. That is a range for T codes, and it is a different statement from our bound $n \leq 38$. Camps-Moreno and coauthors study CSS-T codes, on which T on every qubit is a logical gate, and in 2026 all diagonal transversal gates of a CSS code.

[click] A triorthogonal matrix has even pair and triple overlaps for distinct rows. CCZ needs one odd triple, $K_1 \wedge K_2 \wedge K_3$, so it needs the generalized conditions of Campbell and Howard on the next slide.

Background (not spoken)

- **Odd and even rows.** In a triorthogonal matrix, odd-weight rows give logical X operators and even-weight rows give X-type stabilizers.
- **Triorthogonal space.** Nezami and Haah call a subspace $\mathcal{H}$ triorthogonal if $\sum_j h_j h'_j h''_j \equiv 0 \pmod 2$ for all $h, h', h'' \in \mathcal{H}$, repeats allowed. So every vector has even weight, and the row span of a matrix with odd rows is not such a space. Unital means $\mathbf{1} \in \mathcal{H}$. They classify these spaces and obtain codes by deleting (puncturing) coordinates.
- **CSS-T.** A CSS code on which T on every qubit acts as a logical gate.

Slide 14. Conditions for quasi-transversal CCZ



Clock 14:10–15:40 · **Budget** 90 s · **Clicks** 2

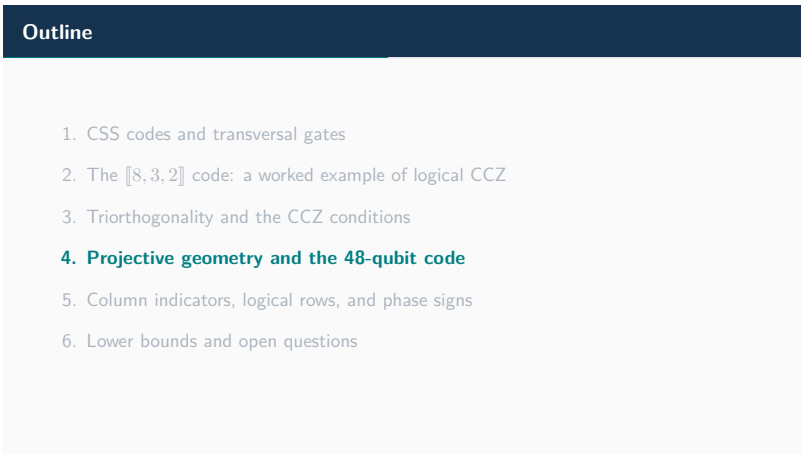
On screen: the nine parity conditions K1–K3, KS1–KS3 and S1–S3.

Reduce the three signed conditions mod 2. Each Γ_j is odd, so a signed weight has the same parity as the ordinary weight. What remains are these nine parity conditions of Campbell and Howard, the CH conditions. K1 to K3 involve only logical rows, KS1 to KS3 mix logical and stabilizer rows, and S1 to S3 involve only stabilizer rows. K3 is the only odd condition, and it selects CCZ.

[click] These nine conditions make T on every qubit, followed by some diagonal Clifford gate C, act as logical CCZ. C is called the correction, and this is quasi-transversal CCZ.

[click] Here is a distinction I want to keep clear. *(pause)* A $T/T^\dagger$ pattern with no correction needs more, namely the signed congruences mod 8 and mod 4. Those are conditions on the signs Γ as well as on the matrix. A code can satisfy all nine parities and still have no valid choice of signs. We will see an example at length 47.

Outline before slide 15 (part 4 of 6)

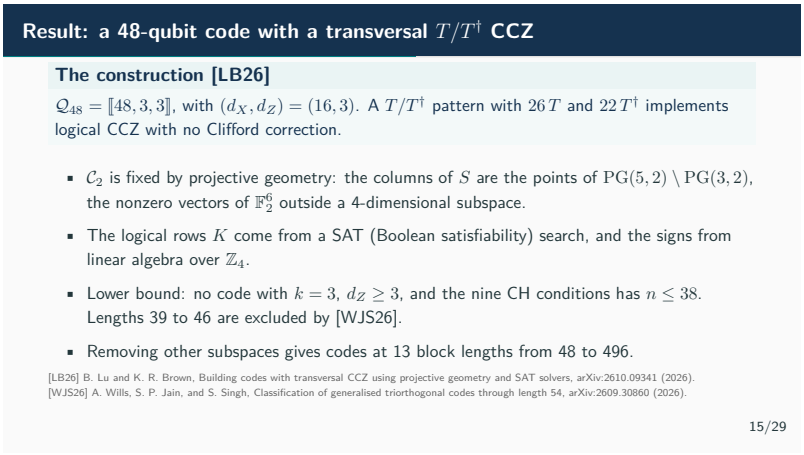


Clock 15:40–15:50 · **Budget** 10 s · **Clicks** 0

On screen: the outline, with part 4 highlighted.

Part four, the construction. We now build the 48-qubit code, starting from the geometry that fixes its stabilizers.

Slide 15. Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

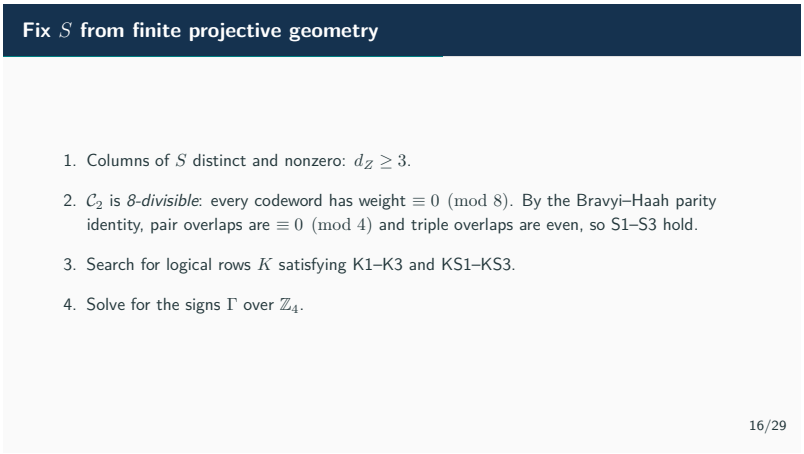


Clock 15:50–17:10 · **Budget** 80 s · **Clicks** 0

On screen: $\mathcal{Q}_{48} = [\![48, 3, 3]\!]$ with $(d_X, d_Z) = (16, 3)$, and four bullets on the construction and the bounds.

Here is the result. $\mathcal{Q}_{48}$ is a $[\![48, 3, 3]\!]$ code, with $d_X = 16$ and $d_Z = 3$. A pattern of 26 T gates and 22 $T^\dagger$ gates implements logical CCZ, with no Clifford correction. It is built in three steps. Geometry fixes the stabilizers: the columns of S are the points of $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$, the nonzero vectors of $\mathbb{F}_2^6$ outside a 4-dimensional subspace. A SAT search, a Boolean satisfiability solver, finds the logical rows K. And linear algebra over $\mathbb{Z}_4$ finds the signs. *(pause)* For the lower bound, our theorem says that no code with three logical qubits, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by the classification of Wills, Jain and Singh. Removing other subspaces gives codes at 13 block lengths from 48 to 496.

Slide 16. Fix S from finite projective geometry



Clock 17:10–18:10 · **Budget** 60 s · **Clicks** 3

On screen: step 1, distinct nonzero columns of S give $d_Z \geq 3$.

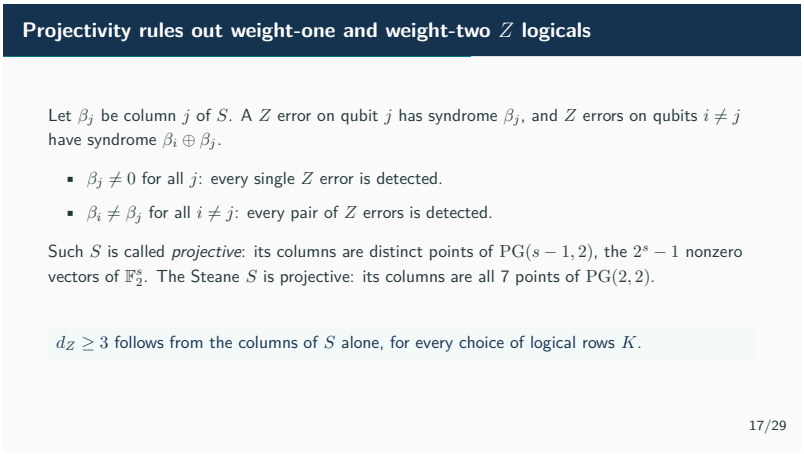
The construction has four steps. First, choose S with distinct nonzero columns. That alone gives $d_Z \geq 3$, as the next slide shows.

[click] Second, require $\mathcal{C}_2$ to be 8-divisible: every codeword has weight divisible by 8. By the Bravyi–Haah parity identity, pair overlaps are then zero mod 4 and triple overlaps are even, so S1 to S3 hold. This divisibility is our design choice. The CH conditions do not require it.

[click] Third, search for logical rows K that satisfy K1 to K3 and KS1 to KS3. This is the only search.

[click] Fourth, solve for the signs Γ over $\mathbb{Z}_4$. That is a linear-algebra test.

Slide 17. Projectivity rules out weight-one and weight-two Z logicals

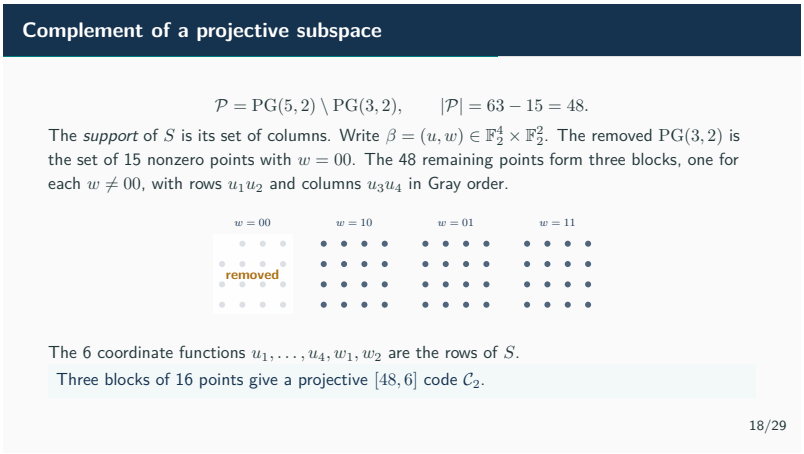


Clock 18:10–19:10 · **Budget** 60 s · **Clicks** 0

On screen: the syndrome of a Z error is a column β_j of S, and distinct nonzero columns detect one or two Z errors.

Why distinct nonzero columns? Let β_j be column j of S. A Z error on qubit j has syndrome β_j, and Z errors on qubits $i \neq j$ have syndrome $\beta_i \oplus \beta_j$. If no column is zero, every single Z error is detected. If the columns are distinct, $\beta_i \oplus \beta_j \neq 0$, so every pair of Z errors is detected too. So $d_Z \geq 3$. *(pause)* Such an S is called projective: its columns are distinct points of $\mathrm{PG}(s-1,2)$, the 2^s-1 nonzero vectors of $\mathbb{F}_2^s$. The Steane S is projective: its columns are all 7 points of $\mathrm{PG}(2,2)$. This is the parity-check argument for a classical code of distance three. It does not involve the logical rows, and it covers d_Z only. We compute d_X for the final code.

Slide 18. Complement of a projective subspace



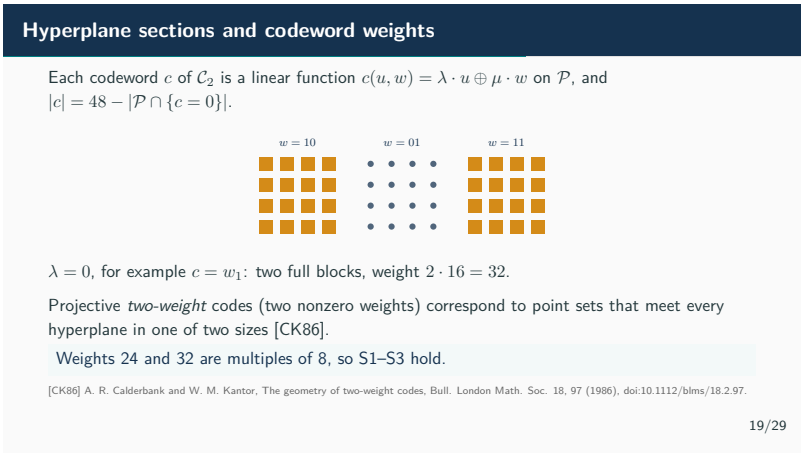
Clock 19:10–20:10 · **Budget** 60 s · **Clicks** 1

On screen: $\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$ with $63 - 15 = 48$ points, drawn as four blocks indexed by w.

Take the 63 points of PG(5, 2) and remove a PG(3, 2), which has 15 points. That leaves 48. The support of S is its set of columns. Write a point as $\beta = (u, w)$, with $u \in \mathbb{F}_2^4$ and $w \in \mathbb{F}_2^2$. The removed subspace is the 15 nonzero points with $w = 00$. In the picture, the four blocks are the values of w. Inside each block the rows are u_1u_2 and the columns are u_3u_4, in Gray order.

[click] Remove the block $w = 00$. The 48 remaining points form three blocks, one for each $w \neq 00$. The six coordinate functions $u_1, \ldots, u_4, w_1, w_2$, evaluated on the 48 points, are the rows of S. Three blocks of 16 points give a projective $[48, 6]$ code, $\mathcal{C}_2$. We keep this picture for the weights, the logical rows and the signs.

Slide 19. Hyperplane sections and codeword weights



Clock 20:10–21:40 · **Budget** 90 s · **Clicks** 1

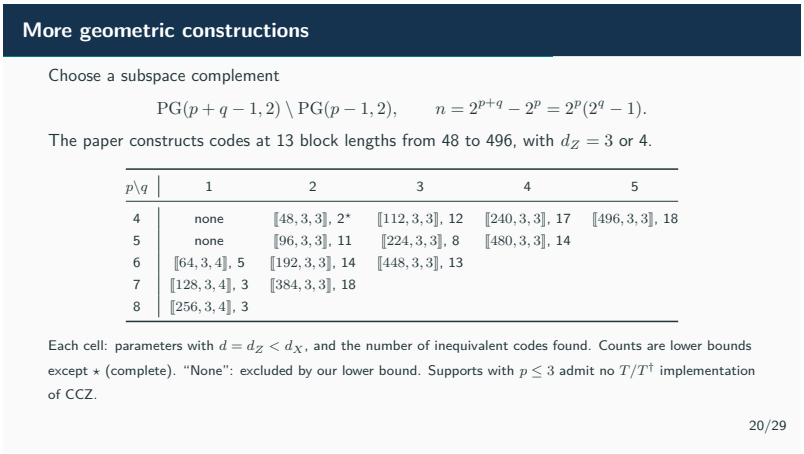
On screen: a codeword of $\mathcal{C}_2$ as a linear function on the 48 points, with u_1 marked in gold.

Now the weights. Each codeword of $\mathcal{C}_2$ is a linear function $c(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on the 48 points. Its weight is 48 minus the number of points on the hyperplane where it vanishes. If $\lambda \neq 0$, for example $c = u_1$, the gold squares show that it is one on eight points of each block. So the weight is $3 \cdot 8 = 24$.

(ask, then wait 7 seconds) What is the weight of w_1? *(Fallback: click and read it off.)*

[click] w_1 is one on two full blocks, so its weight is $2 \cdot 16 = 32$. These are the only two cases. So the nonzero weights are 24 and 32, both multiples of 8, and S1 to S3 hold. *(pause)* A projective code with two nonzero weights is a two-weight code. Calderbank and Kantor show these codes correspond to point sets that meet every hyperplane in one of two sizes.

Slide 20. More geometric constructions



More geometric constructions

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q-1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

$p\backslash q$	1	2	3	4	5
4	none	$[\![48,3,3]\!]$, 2*	$[\![112,3,3]\!]$, 12	$[\![240,3,3]\!]$, 17	$[\![496,3,3]\!]$, 18
5	none	$[\![96,3,3]\!]$, 11	$[\![224,3,3]\!]$, 8	$[\![480,3,3]\!]$, 14	
6	$[\![64,3,4]\!]$, 5	$[\![192,3,3]\!]$, 14	$[\![448,3,3]\!]$, 13		
7	$[\![128,3,4]\!]$, 3	$[\![384,3,3]\!]$, 18			
8	$[\![256,3,4]\!]$, 3				

Each cell: parameters with $d = d_Z < d_X$, and the number of inequivalent codes found. Counts are lower bounds except $\star$ (complete). "None": excluded by our lower bound. Supports with $p \leq 3$ admit no $T/T^\dagger$ implementation of CCZ.

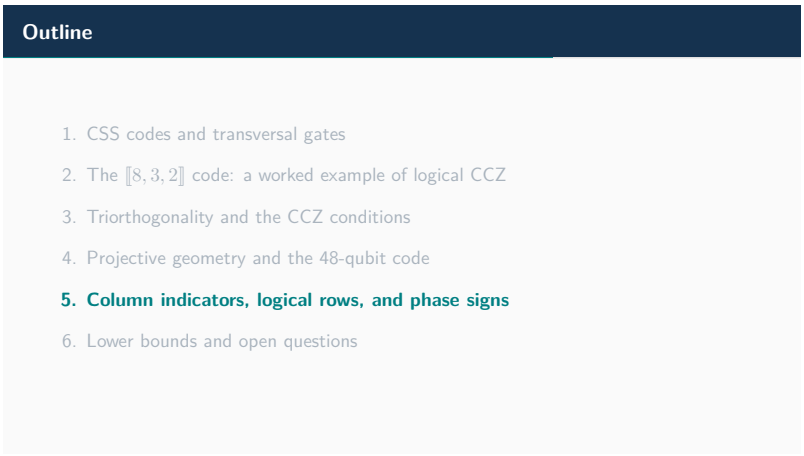
20/29

Clock 21:40–22:40 · **Budget** 60 s · **Clicks** 0

On screen: $n = 2^p(2^q-1)$ for $\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2)$, and a table of codes found for $p = 4$ to 8 and $q = 1$ to 5.

The same idea works for any subspace complement, $\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2)$. The length is $n = 2^p(2^q-1)$. $\mathcal{Q}_{48}$ is the case $p = 4$, $q = 2$. The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4. Each cell gives the parameters and the number of inequivalent codes we found. For example, $p = 6$ and $q = 1$ gives 64 qubits with $d_Z = 4$. *(pause)* Two qualifications. Only the starred entry at 48 is a complete census for that support. The other counts are lower bounds. The entries marked "none" are excluded by our lower bound, and supports with $p \leq 3$ admit no $T/T^\dagger$ implementation of CCZ. This is a family of finite constructions, not an asymptotic statement.

Outline before slide 21 (part 5 of 6)

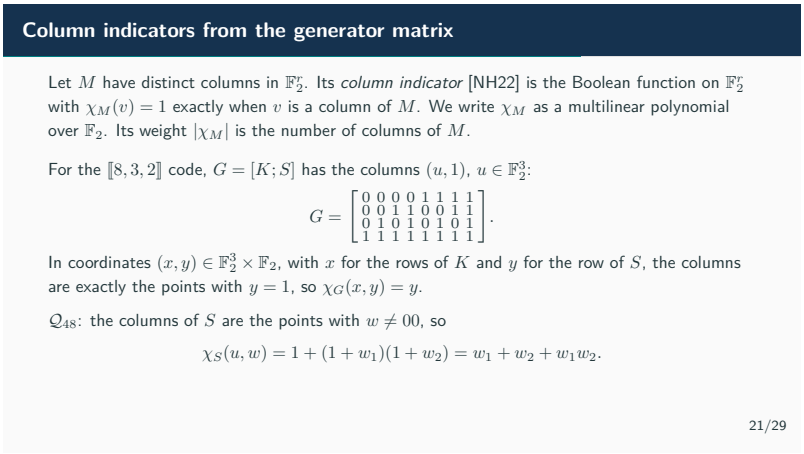


Clock 22:40–22:50 · **Budget** 10 s · **Clicks** 0

On screen: the outline, with part 5 highlighted.

Part five. Geometry has fixed the stabilizers. Now we find the logical rows using column indicators, and then solve for the signs.

Slide 21. Column indicators from the generator matrix



Clock 22:50–24:10 · **Budget** 80 s · **Clicks** 2

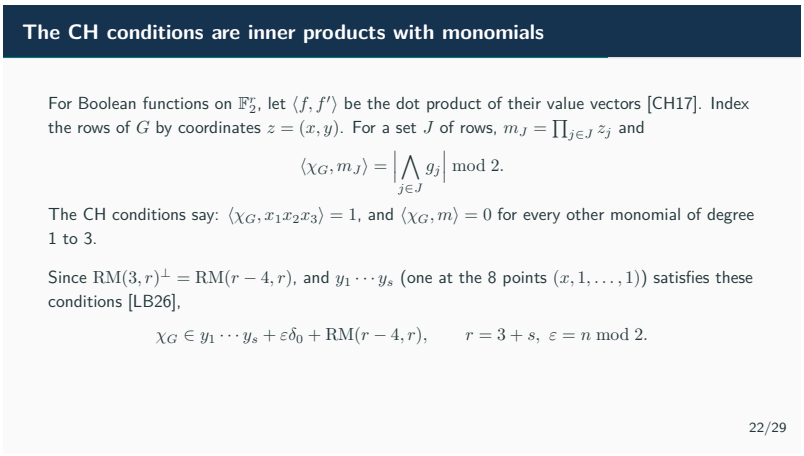
On screen: the definition of the column indicator χ_M.

To find the logical rows we switch from rows to columns. Let M have distinct columns in $\mathbb{F}_2^r$. Its column indicator, as in Nezami and Haah, is the Boolean function χ_M on $\mathbb{F}_2^r$ with $\chi_M(v) = 1$ exactly when v is a column of M. We write it as a multilinear polynomial over $\mathbb{F}_2$. Its weight is the number of columns of M.

[click] For the $[\![8, 3, 2]\!]$ code, $G = [K; S]$ has the columns $(u, 1)$ for every $u \in \mathbb{F}_2^3$. In coordinates (x, y), with x for the rows of K and y for the row of S, every x occurs with $y = 1$, and no column has $y = 0$. So $\chi_G(x, y) = y$.

[click] For $\mathcal{Q}_{48}$, the columns of S are the points with $w \neq 00$. So $\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2$. It has degree two and does not depend on u.

Slide 22. The CH conditions are inner products with monomials



Clock 24:10–25:40 · **Budget** 90 s · **Clicks** 2

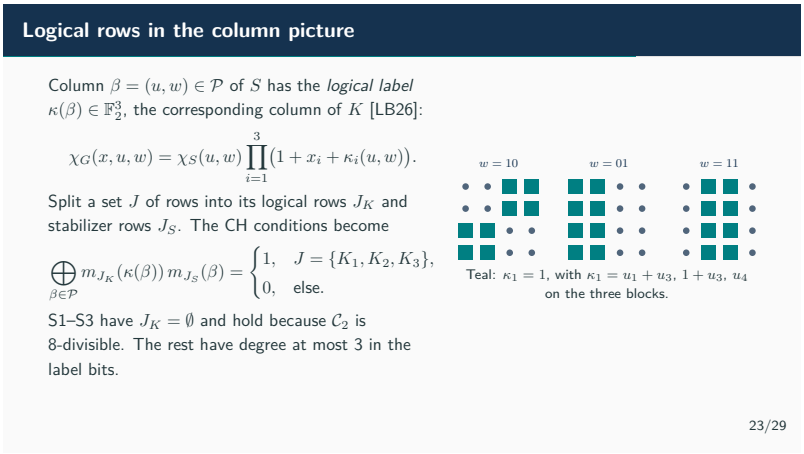
On screen: the inner product $\langle f, f' \rangle$ and the identity $\langle \chi_G, m_J \rangle = |\bigwedge_{j \in J} g_j| \bmod 2$.

Why columns? Because overlaps become inner products. For Boolean functions on $\mathbb{F}_2^r$, $\langle f, f' \rangle$ is the dot product of their value vectors, mod 2. Index the rows of G by coordinates $z = (x, y)$. For a set J of rows, let $m_J = \prod_{j \in J} z_j$ be the monomial. Then $\langle \chi_G, m_J \rangle$ is the weight of the overlap of those rows, mod 2. Summing over the columns is summing over the points where χ_G is one.

[click] So the CH conditions say this: $\langle \chi_G, x_1 x_2 x_3 \rangle = 1$, and $\langle \chi_G, m \rangle = 0$ for every other monomial of degree one to three.

[click] The dual of $\mathrm{RM}(3, r)$ is $\mathrm{RM}(r-4, r)$. The product $y_1 \cdots y_s$ satisfies these conditions. It is one at the eight points $(x, 1, \ldots, 1)$, and among these monomials only $x_1 x_2 x_3$ has odd inner product with it. So every valid χ_G lies in $y_1 \cdots y_s + \varepsilon\delta_0 + \mathrm{RM}(r-4, r)$, where $r = 3 + s$. Here δ_0 is the indicator of the origin and $\varepsilon = n \bmod 2$. That coset statement is a lemma in our paper.

Slide 23. Logical rows in the column picture



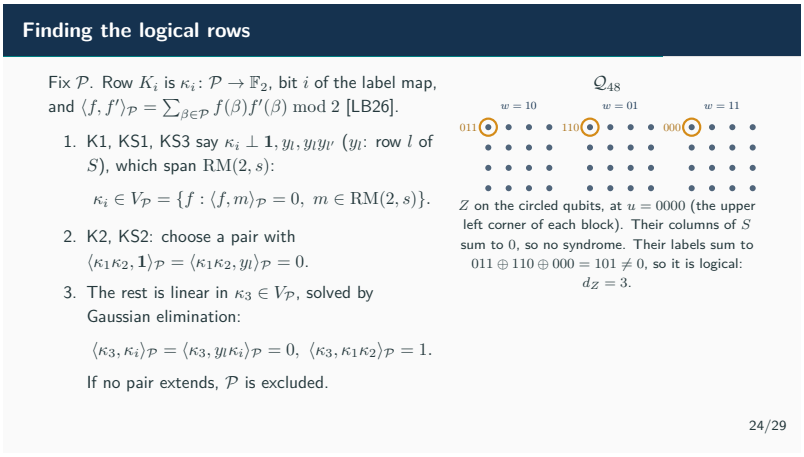
Clock 25:40–27:10 · **Budget** 90 s · **Clicks** 1

On screen: the logical label $\kappa(\beta)$, the factorization of χ_G, and the CH conditions as sums over the 48 points.

For $\mathcal{Q}_{48}$, S is fixed, so the unknowns are the logical rows. In the column picture, each column β of S has a logical label $\kappa(\beta) \in \mathbb{F}_2^3$, the corresponding column of K. Then $\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3}(1+x_i+\kappa_i(u, w))$, which forces x to equal κ at each point. Split a set J of rows into its logical rows J_K and stabilizer rows J_S. The CH conditions become a sum over the 48 points, which must equal one exactly for the logical triple. S1 to S3 have no logical rows, and they hold because $\mathcal{C}_2$ is 8-divisible. The rest are equations of degree at most three in the label bits, three per point.

[click] Here is the first logical row of $\mathcal{Q}_{48}$ on the support. κ_1 is $u_1 + u_3$ on the first block, $1 + u_3$ on the second, and u_4 on the third. It has eight ones in each block.

Slide 24. Finding the logical rows



Clock 27:10–29:10 · **Budget** 120 s · **Clicks** 3

On screen: step 1 of Algorithm 1, $\kappa_i \in V_{\mathcal{P}}$, and on the right the support of $\mathcal{Q}_{48}$ with $\dim V_{\mathcal{P}} = 27$. The clicks show κ_1 and κ_2, then κ_3, then the weight-3 Z logical with the labels of its three qubits.

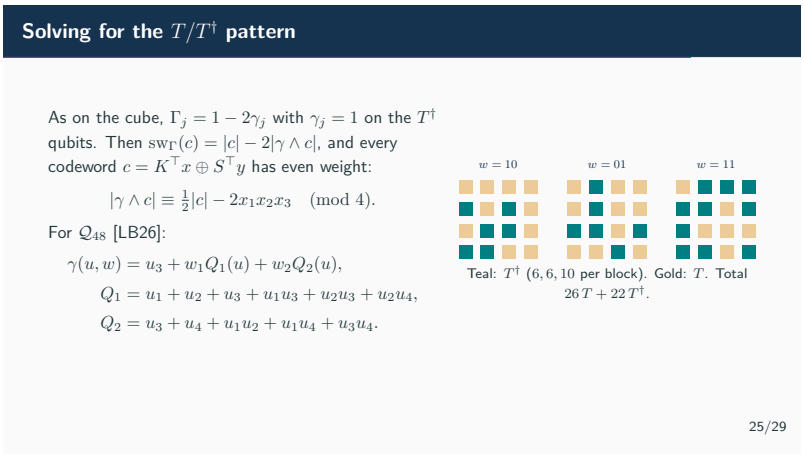
This is Algorithm 1 of the paper. Fix the support $\mathcal{P}$. Row K_i, read as a function on the points of $\mathcal{P}$, is κ_i, the i-th bit of the label map. So $\kappa_1(\beta) = 1$ means K_1 has a one at qubit β. Step one: K1, KS1 and KS3 for a single row say that κ_i is orthogonal to $\mathbf{1}$, to each y_l, and to each product $y_l y_{l'}$. Those functions span $\mathrm{RM}(2, s)$. So κ_i lies in the linear subspace $V_{\mathcal{P}}$. *(pause)* On the right is the support of $\mathcal{Q}_{48}$, where $V_{\mathcal{P}}$ has dimension 27. Its logical rows, found by our SAT search, pass each step.

[click] Step two: choose a pair $\kappa_1, \kappa_2 \in V_{\mathcal{P}}$ whose product is orthogonal to $\mathbf{1}$ and to every y_l. Those are the conditions K2 and KS2. A square is a qubit where K_1 or K_2 has a one. Teal means only K_1, gold means only K_2, and red means both. The red squares are the overlap $K_1 \wedge K_2$. There are 12, an even number, as K2 requires.

[click] Step three: for a fixed pair, the conditions on κ_3 are linear. Gaussian elimination over $\mathbb{F}_2$ finds κ_3 or proves that none exists. The squares now show where K_3 has a one, and the red ones are where K_1 and K_2 also do, the overlap $K_1 \wedge K_2 \wedge K_3$. There are 7, an odd number, as K3 requires.

[click] If no pair extends, the support is excluded. *(pause)* Now the distance. Take the three circled qubits at $u = 0000$, the upper left corner of each block. Their columns of S sum to zero, so Z on them has no syndrome. Their labels, the columns of K printed beside the circles, sum to 101, not zero. So this Z flips the sign of $x_1 \oplus x_3$. It is a logical operator of weight three, like the Fano line on the Steane code, and with projectivity $d_Z = 3$.

Slide 25. Solving for the $T/T^\dagger$ pattern



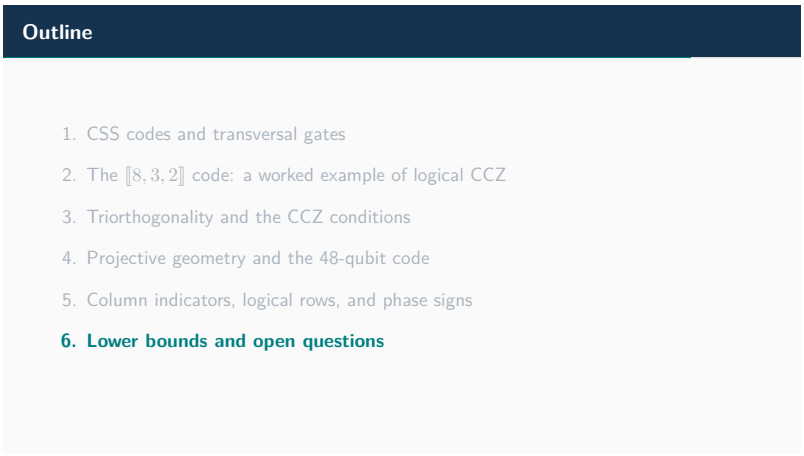
Clock 29:10–30:40 · **Budget** 90 s · **Clicks** 1

On screen: $\Gamma_j = 1 - 2\gamma_j$ and the congruence $|\gamma \wedge c| \equiv \frac{1}{2}|c| - 2x_1x_2x_3 \pmod 4$.

The last step is the signs. As on the cube, write $\Gamma_j = 1 - 2\gamma_j$, with γ_j a bit that is one where we put $T^\dagger$. Then $\mathrm{sw}_\Gamma(c) = |c| - 2|\gamma \wedge c|$, an ordinary integer count. Every codeword has even weight, so we can divide the gate condition by two. The weight of the overlap must satisfy $|\gamma \wedge c| \equiv \frac{1}{2}|c| - 2x_1x_2x_3 \pmod 4$. That is linear in γ over $\mathbb{Z}_4$, with the extra requirement that γ is binary. Elimination over $\mathbb{Z}_4$ decides whether a solution exists.

[click] For $\mathcal{Q}_{48}$, here is the solution from the paper. $\gamma(u, w) = u_3 + w_1Q_1(u) + w_2Q_2(u)$, with two quadratic functions Q_1 and Q_2. In the picture, teal is $T^\dagger$, with 6, 6 and 10 in the three blocks, 22 in all. Gold is T, 26 in all. These are coordinate positions, not a hardware layout.

Outline before slide 26 (part 6 of 6)

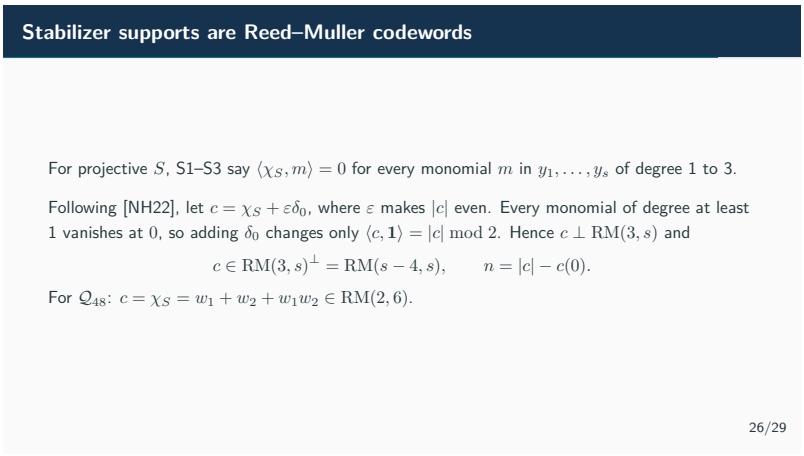


Clock 30:40–30:50 · **Budget** 10 s · **Clicks** 0

On screen: the outline, with part 6 highlighted.

Part six, the last part. The construction is complete. Now the same Boolean view explains the lower bound and what remains open.

Slide 26. Stabilizer supports are Reed–Muller codewords

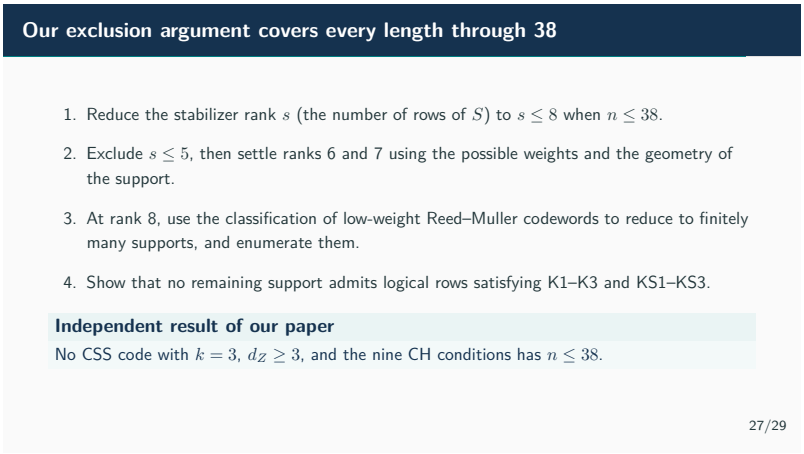


Clock 30:50–32:40 · **Budget** 110 s · **Clicks** 0

On screen: S1–S3 as $\langle \chi_S, m \rangle = 0$ for monomials of degree 1 to 3, and $c = \chi_S + \varepsilon\delta_0 \in \mathrm{RM}(s-4, s)$.

Why can no short code work? Take a projective S. In the column picture, S1 to S3 say that χ_S has inner product zero with every monomial of degree one to three. Following Nezami and Haah, let $c = \chi_S + \varepsilon\delta_0$, with ε chosen to make the weight of c even. Every monomial of positive degree vanishes at the origin. So adding δ_0 changes only the inner product with $\mathbf{1}$, which is the parity of the weight. *(pause)* Hence $c \perp \mathrm{RM}(3, s)$, and $c \in \mathrm{RM}(s-4, s)$. The block length is $n = |c| - c(0)$. For $\mathcal{Q}_{48}$, $c = w_1 + w_2 + w_1 w_2$, a codeword of $\mathrm{RM}(2, 6)$. So a short code needs a low-weight codeword of a Reed–Muller code. Low-weight codewords of Reed–Muller codes are classified, and that is what makes the search below length 39 finite.

Slide 27. Our exclusion argument covers every length through 38

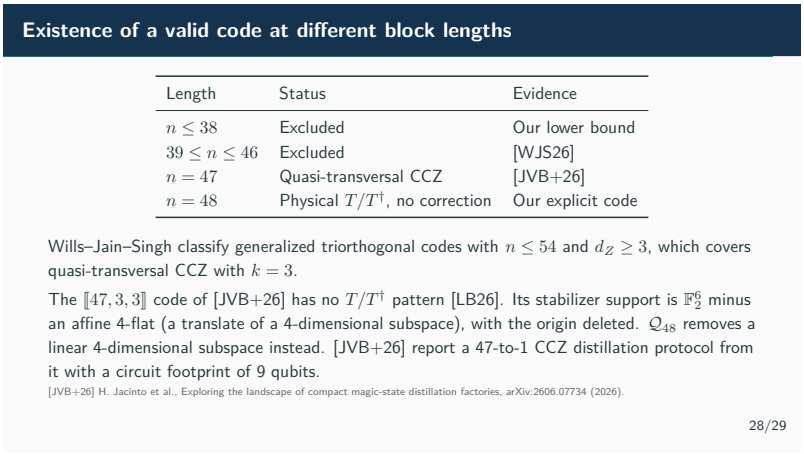


Clock 32:40–34:10 · **Budget** 90 s · **Clicks** 0

On screen: four proof stages, and the theorem: no CSS code with $k = 3$, $d_Z \geq 3$ and the nine CH conditions has $n \leq 38$.

The proof that nothing works through length 38 has four stages. First, reduce the stabilizer rank s, the number of rows of S, to $s \leq 8$ when $n \leq 38$. Second, exclude $s \leq 5$, then settle ranks 6 and 7 using the weight distribution and the geometry of the support. Third, at rank 8, use the classification of low-weight Reed–Muller codewords to reduce to finitely many supports, and enumerate them. Fourth, show that no remaining support admits logical rows satisfying K1 to K3 and KS1 to KS3. *(pause)* The result is our theorem. No CSS code with $k = 3$, $d_Z \geq 3$ and the nine CH conditions has $n \leq 38$. Two remarks. The rank-8 coverage and the logical-row exclusions are complete, not sampled. And because the theorem assumes only the nine parity conditions, it rules out quasi-transversal codes as well as $T/T^\dagger$ patterns.

Slide 28. Existence of a valid code at different block lengths



Existence of a valid code at different block lengths

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify generalized triorthogonal codes with $n \leq 54$ and $d_Z \geq 3$, which covers quasi-transversal CCZ with $k = 3$.

The $[\![47, 3, 3]\!]$ code of [JVB+26] has no $T/T^\dagger$ pattern [LB26]. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat (a translate of a 4-dimensional subspace), with the origin deleted. $\mathcal{Q}_{48}$ removes a linear 4-dimensional subspace instead. [JVB+26] report a 47-to-1 CCZ distillation protocol from it with a circuit footprint of 9 qubits.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

28/29

Clock 34:10–36:10 · **Budget** 120 s · **Clicks** 0

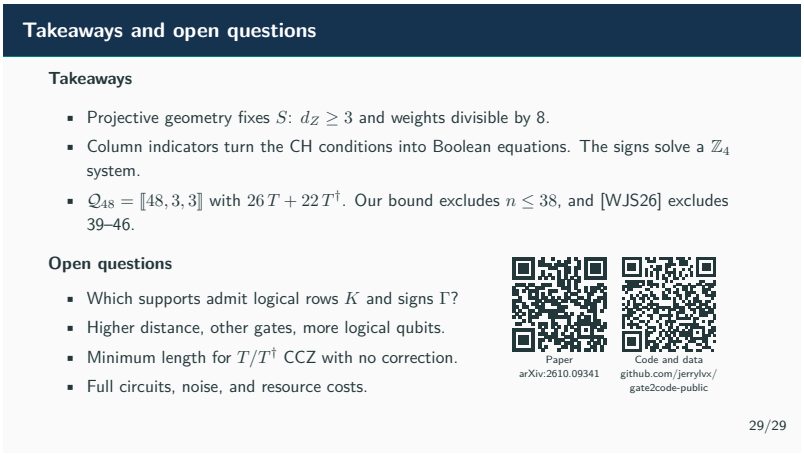
On screen: a table of lengths, status and evidence, and three sentences on the $[\![47, 3, 3]\!]$ code.

Here is the situation by length. Up to 38 is excluded by our bound. 39 to 46 is excluded by Wills, Jain and Singh, who classify generalized triorthogonal codes up to length 54 with $d_Z \geq 3$. That classification covers quasi-transversal CCZ with three logical qubits. At 47 there is a quasi-transversal CCZ code, the $[\![47, 3, 3]\!]$ code of Jacinto and coauthors. At 48 we have $\mathcal{Q}_{48}$, with a $T/T^\dagger$ pattern and no correction. *(pause)* Now the distinction between 47 and 48. The 47-qubit code satisfies the nine parities, but it has no $T/T^\dagger$ pattern. We show this in our paper. So at 47 a Clifford correction is needed. That statement is about this specific code. It does not show that 48 is the minimum length for a $T/T^\dagger$ pattern, and it does not show that the 47-qubit code is unique. It is also not $\mathcal{Q}_{48}$ with one point removed. Its stabilizer support is $\mathbb{F}_2^6$ minus an affine 4-flat, a translate of a 4-dimensional subspace, with the origin deleted. $\mathcal{Q}_{48}$ removes a linear 4-dimensional subspace. Finally, Jacinto and coauthors report a 47-to-1 CCZ distillation protocol from their code with circuit footprint of 9 qubits. That number is theirs, and it is about the circuit, not the code length.

Background (not spoken)

- **If asked why the 47-qubit code is not $\mathcal{Q}_{48}$ punctured.** The stabilizer weights differ: 16, 24 and 32 for the 47-qubit code, and 23, 24, 31 and 32 for any puncture of $\mathcal{Q}_{48}$.

Slide 29. Takeaways and open questions

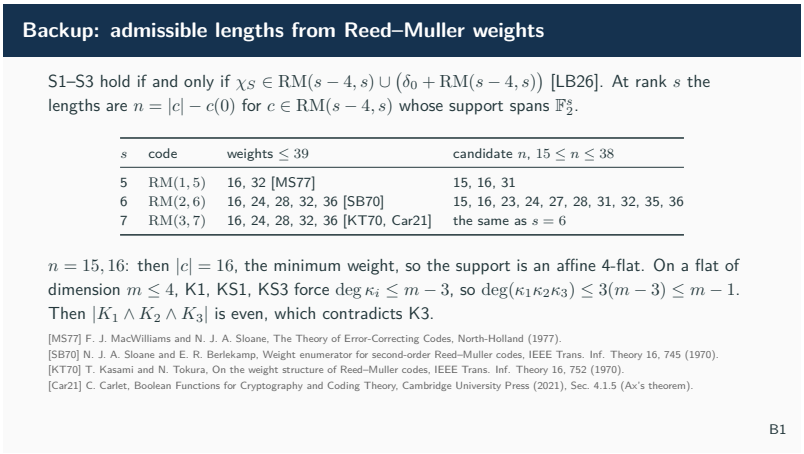


Clock 36:10–37:40 · **Budget** 90 s · **Clicks** 0

On screen: three takeaways, four open questions, and QR codes for the paper and the code.

Here is what we contributed. *(pause)* Projective geometry fixes S, giving $d_Z \geq 3$ and weights divisible by eight. Column indicators turn the CH conditions into Boolean equations for the logical rows, and the signs solve a linear system over $\mathbb{Z}_4$. Together these give $\mathcal{Q}_{48}$, a $[\![48, 3, 3]\!]$ code in which 26 T and 22 $T^\dagger$ gates act as logical CCZ. Our bound excludes lengths up to 38, and Wills, Jain and Singh exclude 39 to 46. In one sentence: geometry fixes the stabilizers, a search finds the logical rows, and linear algebra over $\mathbb{Z}_4$ finds the signs. *(pause)* Some open questions. Which supports admit logical rows and signs? Can we reach higher distance, other gates, or more logical qubits? What is the minimum length for a CCZ pattern of T and $T^\dagger$ with no correction? And what do full circuits cost once noise is included? If you work on divisible codes, or on completing a code from a fixed support, I would like to hear your ideas. The QR codes link to the paper and the code. *(salute, then stop)* It has been a pleasure to present this work to algebraists and coding theorists here at Virginia Tech. I look forward to your questions.

Backup B1. Backup: admissible lengths from Reed–Muller weights



Backup: admissible lengths from Reed–Muller weights

S1–S3 hold if and only if $\chi_S \in \mathrm{RM}(s-4,s) \cup \big(\delta_0 + \mathrm{RM}(s-4,s)\big)$ [LB26]. At rank s the lengths are $n = |c| - c(0)$ for $c \in \mathrm{RM}(s-4,s)$ whose support spans $\mathbb{F}_2^s$.

s	code	weights ≤ 39	candidate n, $15 \leq n \leq 38$
5	$\mathrm{RM}(1,5)$	16, 32 [MS77]	15, 16, 31
6	$\mathrm{RM}(2,6)$	16, 24, 28, 32, 36 [SB70]	15, 16, 23, 24, 27, 28, 31, 32, 35, 36
7	$\mathrm{RM}(3,7)$	16, 24, 28, 32, 36 [KT70, Car21]	the same as $s = 6$

$n = 15, 16$: then $|c| = 16$, the minimum weight, so the support is an affine 4-flat. On a flat of dimension $m \leq 4$, K1, KS1, KS3 force $\deg \kappa_i \leq m-3$, so $\deg(\kappa_1\kappa_2\kappa_3) \leq 3(m-3) \leq m-1$. Then $|K_1 \wedge K_2 \wedge K_3|$ is even, which contradicts K3.

[MS77] F. J. MacWilliams and N. J. A. Sloane, The Theory of Error-Correcting Codes, North-Holland (1977).
[SB70] N. J. A. Sloane and E. R. Berlekamp, Weight enumerator for second-order Reed–Muller codes, IEEE Trans. Inf. Theory 16, 745 (1970).
[KT70] T. Kasami and N. Tokura, On the weight structure of Reed–Muller codes, IEEE Trans. Inf. Theory 16, 752 (1970).
[Car21] C. Carlet, Boolean Functions for Cryptography and Coding Theory, Cambridge University Press (2021), Sec. 4.1.5 (Ax's theorem).

B1

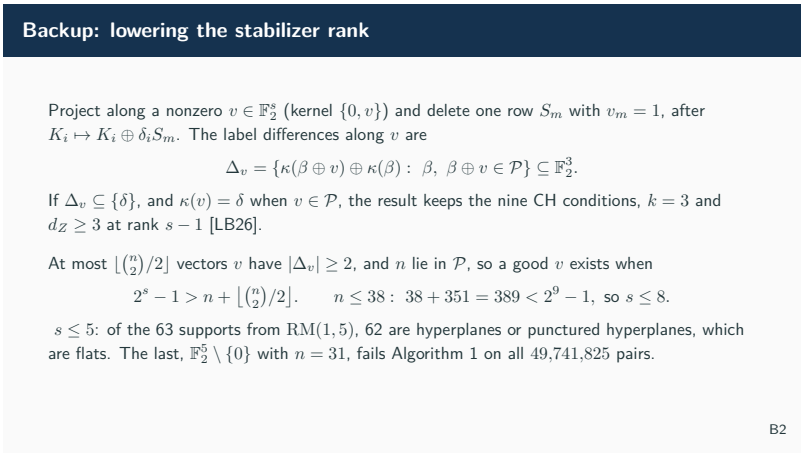
Clock questions only · **Clicks** 1

On screen: the admissible-length theorem and a table of Reed–Muller weights for $s = 5, 6, 7$. The click adds the affine-flat argument for $n = 15, 16$.

Use this if asked why the bound reduces to Reed–Muller weights. The conditions S1 to S3 hold exactly when χ_S, possibly after adding δ_0, is a codeword c of $\mathrm{RM}(s-4, s)$, and the length is $n = |c| - c(0)$. So every admissible length comes from a Reed–Muller weight. At ranks 5, 6 and 7 the weights up to 39 are known: from first-order codes, from the Sloane–Berlekamp enumerator, and from Kasami–Tokura together with Ax's theorem. That leaves the candidate lengths in the table.

[click] The two smallest candidates, 15 and 16, come from the minimum weight 16, whose support is an affine 4-flat. On such a flat the single-row conditions force each label κ_i to have low degree, so their product has degree below the dimension, and the triple overlap is even. That contradicts K3.

Backup B2. Backup: lowering the stabilizer rank



Clock questions only · **Clicks** 2

On screen: the projection along v, the label differences Δ_v, and the condition that keeps the CH conditions at rank $s - 1$. The clicks add the counting bound, then the case $s \leq 5$.

Use this if asked why only ranks 6, 7 and 8 need a search. Project the columns along a nonzero vector v and delete one stabilizer row. Columns that differ by v merge. If all such pairs have the same label difference δ, the merged columns carry equal labels, and the smaller code keeps the nine CH conditions, three logical qubits and $d_Z \geq 3$.

[click] Each pair of support points determines one difference v. So at most $\lfloor \binom{n}{2}/2 \rfloor$ vectors have two or more label differences, and n vectors lie in the support. If $2^s - 1$ exceeds their sum, a good v exists. For $n \leq 38$ that holds whenever $s \geq 9$, so $s \leq 8$.

[click] At rank 5 or less, every support but one is a flat, and the last one, all 31 nonzero points of $\mathbb{F}_2^5$, fails Algorithm 1 on every pair.

Backup B3. Backup: the finite cases at ranks 6, 7 and 8

Backup: the finite cases at ranks 6, 7 and 8

s	$\lvert c\rvert$	cases	result
6, 7	16	affine 4-flat	K3 fails
6	24–36	six affine classes (Dickson forms). The degree-one class lies in a hyperplane, so the rank drops to 5	no κ_3
7	24, 28	1 and 2 classes [KT70]	no κ_3
7	32	$y_1 y_2$ (in a hyperplane) and 3 cubic classes [KTA76]	no κ_3
7	36	9,792 functions, 7,038 supports	no κ_3
8	≤ 30	counting: $n + \lfloor \binom{n}{2}/2 \rfloor \leq 247 < 2^8 - 1$	rank drops
8	32, $\deg c \leq 3$	affine 5-flat, rank ≤ 6	excluded
8	32–38, $\deg c = 4$	$c = y_7 f + y_8 g + y_7 y_8 h$ with $\lvert c\rvert = \lvert f\rvert + \lvert g\rvert + \lvert f + g + h\rvert$, $(\lvert f\rvert, \lvert g\rvert) \in \{(8,8), (8,12), (12,12), (8,14)\}$, 2,399 orbit representatives of g, 469,728 supports	no κ_3

Every case ends in a flat or hyperplane reduction, or in Algorithm 1 finding no third logical row.

[KTA76] T. Kasami, N. Tokura, and S. Azumi, On the weight enumeration of weights less than 2.5d of Reed–Muller codes, Inf. Control 30, 380 (1976).

B3

Clock questions only · **Clicks** 0

On screen: a table of the cases at ranks 6, 7 and 8 and how each is excluded.

Use this if asked what the finite search covers. At rank 6, the Dickson forms leave six classes, and none admits a third logical row. At rank 7, the weights 24, 28 and 32 leave a few classes from Kasami–Tokura and Kasami–Tokura–Azumi, and weight 36 is enumerated directly. At rank 8, small weights reduce the rank by counting, and degree three forces an affine flat. The remaining degree-4 polynomials split as $y_7 f + y_8 g + y_7 y_8 h$, and the weight bound leaves four weight pairs for f and g. We list orbit representatives of g, enumerate h, and run Algorithm 1 on every resulting support. Every case ends in a flat or hyperplane reduction, or in no third logical row.

Backup B4. Backup: dictionary with [CLM+26]

Backup: dictionary with [CLM+26]

[CLM+26]	This talk
$U(b) = \bigotimes_j \mathrm{diag}(1, \omega^{b_j})$, $\omega = e^{2\pi i/N}$, $N = 2^\ell$	$N = 8$, $b \equiv \Gamma$: $T = U(1)$, $T^\dagger = U(-1)$
$b \cdot w$ for $w \in \mathbb{F}_2^n$	$\mathrm{sw}_\Gamma(w)$
$U(b)$ fixes the code iff	rows with a stabilizer row: $\mathrm{sw}_\Gamma(S_l) \equiv 0$ (8),
$b \in H_N = \left(\beta_2 \star \bigcup_{i<\ell} 2^i \beta_1^{(i)}\right)^{\perp_N}$	$\mathrm{sw}_\Gamma(S_l \wedge g) \equiv 0$ (4), $\mathrm{sw}_\Gamma(S_l \wedge g \wedge g') \equiv 0$ (2)
logical action $\omega^{b\cdot(Bv)} = \prod_J C_J\text{-}U(a_J)$,	CCZ $= C_{123}$-$U(4)$: $\mathrm{sw}_\Gamma(K_i) \equiv 0$ (8),
$a_J = (-2)^{\lvert J\rvert - 1}\, b \cdot (w_{j_1} \star \cdots \star w_{j_h})$	$\mathrm{sw}_\Gamma(K_i \wedge K_{i'}) \equiv 0$ (4),
	$\mathrm{sw}_\Gamma(K_1 \wedge K_2 \wedge K_3) \equiv 1$ (2)

Our signed conditions are their H_8 with $b \in \{\pm 1\}^n$ and the logical action fixed to CCZ. We solve the converse problem: given the action, find S, K and b.

B4

Clock questions only · **Clicks** 0

On screen: a two-column table matching the notation of Camps-Moreno et al. 2026 with the signed conditions of this talk.

Use this if asked how this relates to the transversal-gates framework of Camps-Moreno, López, Matthews, Rengaswamy and San José. They write a diagonal transversal gate as $U(b)$ with phases ω^{b_j}, ω a 2^ℓ-th root of unity. Our $T/T^\dagger$ pattern is the case $N = 8$ with $b \equiv \Gamma$, and their $b \cdot w$ is our signed weight. Their group H_N of gates that fix the code gives our conditions on overlaps that include a stabilizer row. Their expansion of the logical action into multi-controlled phases gives our conditions on the logical rows, once we ask for exactly CCZ. So our signed conditions are their H_8 with $b \in \{\pm 1\}^n$ and the logical action fixed to CCZ. Our work is the converse problem: given the action, find S, K and b.

Background (not spoken)

- **Their notation.** Bases $\beta_2 \subseteq \beta_1$ of $\mathcal{C}_2 \subseteq \mathcal{C}_1$ are our rows of S and of G, and the logical basis vectors w_j are our rows K_j. $A^{(r)}$ is the set of products of r distinct elements of A.
- **Coefficient.** Their proposition on controlled phases writes a_J with $\mathrm{wt}(b \star w_{j_1} \star \cdots)$, where wt is the Hamming weight. That equals $b \cdot (w_{j_1} \star \cdots)$ when b is a 0/1 vector. Their lemma on the logical action, $\omega^{b\cdot(Bv)}$, gives the dot product in general, which is the reading needed when $T^\dagger$ entries are 7.
- **Their "transversal logical".** They call $U(b)$ a transversal logical when its logical action is a tensor product of single-qubit gates. Logical CCZ is not of that form, so our gate lies in H_8 but not in their T_8.

After the talk

Leave slide 29 on screen during questions. Backup slides B1 to B4 follow it: Reed–Muller lengths, rank reduction, the finite cases, and the dictionary with Camps-Moreno et al. Prepared answers to likely technical questions are in questions.md.